Sustainable Finance & Fund Structuring

A Study of French Private Equity

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Foreword & acknowledgments

Disclaimer

The analyses, interpretations, and conclusions presented in this document are solely those of the author. They should not be interpreted as representing the official position of the institutions in which I work or have worked, of the project's partners, or of its funders. Any error, omission, or inaccuracy remains my sole responsibility.

This study is in no way normative and does not conclude that one type of structuring is "superior" to another. Its main purpose is to provide a transparent and independent benchmark of private equity fund structuring practices around sustainable finance issues.

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Glossary

GPs / General Partners: management companies that raise and manage funds.

LPs / Limited Partners: investors providing capital to the fund.

SFDR: European framework for the classification and transparency of sustainable financial products.

ESG: environmental, social, and governance issues.

Sustainable investment objective: explicit social and/or environmental objective pursued by the vehicle.

IMM: Impact Management & Measurement mechanism used to define, monitor, and measure sustainability objectives.

EF Committee: extra-financial committee responsible for monitoring, validating, or framing certain sustainable parameters of the fund.

EF Carried: carried interest indexed to extra-financial criteria.

ESG contractualization: degree to which ESG or sustainable commitments are embedded in the formal documentation of the fund.

Financial additionality: allocation of capital to activities or solutions directly contributing to a sustainable objective.

Operational additionality: transformation of portfolio companies through support, governance, and shareholder engagement.

Macro cluster: grouping of funds according to their overall level of sustainable structuring.

Modality: the concrete form taken by a mechanism when it exists.

Fund organizational architecture: combination of objectives, resources, processes, incentive mechanisms, and control mechanisms.

Organizational signatures: typical fund-structuring configurations identified by the analysis.

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Executive summary

Sustainable finance & private equity: why this study?

Sustainable finance, understood as the integration of sustainability criteria into investment decisions both to better manage the financial risks related to sustainability issues and to generate positive social or environmental impact through allocation, has emerged as a structural transformation of the financial system. This has occurred under the combined effect of long-term macroeconomic constraints, regulatory pressures, and growing expectations from clients. Planetary boundaries, the financing needs associated with the Sustainable Development Goals, the just transition, and rising social and climate risks are pushing financial actors to integrate sustainability issues more explicitly into their decisions. This dynamic has been underway for twenty years, and the PRI now bring together 5,300 signatories representing $128.3 trillion in assets (UNPRI, 2025). At the same time, the impact investing market has grown strongly in recent years, reaching $1.571 trillion in 2024 (+21.1% p.a. since 2019, GIIN).

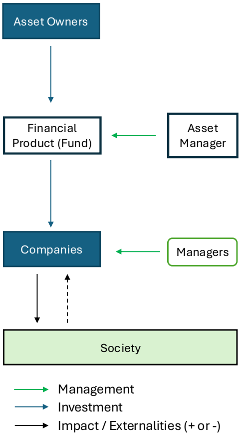
Private equity is also affected by this transformation. Historical investment strategies are evolving to integrate ESG risks, and GPs are developing new investment theses to capture growth opportunities linked to financing the SDGs, particularly on climate-related issues. GPs highlight the advantages of the private markets with regard to sustainability issues: long-term support, proximity to management teams, operational transformation capacity, and access to assets providing innovative solutions to climate and social challenges. In this context, **69%** of GPs view ESG as a long-term value-creation lever (according to a recent PwC survey, 2024), fundraising committed to sustainability increased by approximately **30% per year between 2015 and 2020** (McKinsey survey, 2021), and the private markets already accounted for **65%** of impact assets in 2022 (GIIN, 2023). In France, for example, cleantech and energy-transition infrastructure investments reached **EUR 2.4bn across 150 transactions** in 2024 (France Invest, 2025).

Despite this rapid develoment, the systemic impact of these practices remains uncertain (Schlütter et al., 2023). That impact depends on the systematic activation of the concrete levers of sustainable finance, in particular capital allocation that integrates sustainability factors and shareholder engagement on sustainability issues. In an intermediated financial system such as ours, and even more so in private equity, that activation itself depends on the nature of the mandates entrusted to managers and on their organizational implications: assigned objectives, dedicated resources, and incentive and control mechanisms. Yet these mandates, their translation into the organizational architecture of funds, and the practices actually implemented remain structurally little visible outside the fund's legal documentation, which is not public by nature. **This study aims precisely to document and make visible the detail of these mandates and their organizational implications**, and to discuss their characteristics in relation to the funds' sustainable finance strategies.

This issue is becoming even more pressing in a context of private-market *"retailization"*: at the end of 2024 in France, fundraising directed at retail investors was still increasing by **29%**, reaching EUR 2.4bn (France Invest, 2025).

A blind spot: the upstream structuring of investment organizations

Financial system structure



Existing academic and practitioner work (TCFD, PRI, iCI, France Invest, Impact Frontiers, etc.) mostly focuses on the downstream part of the investment chain. It analyzes interactions between investors and portfolio companies: capital-allocation strategies (positive or negative screening, impact investing, *blended finance*), shareholder-engagement practices and portfolio-company support, as well as their effects on company behavior and, ultimately, on society. This body of work enables the identification and formulation of recommendations on the main levers that GPs can activate to integrate sustainability issues and potentially generate positive social or environmental impact.

This study focuses on an upstream phase: the organizational structuring of investment vehicles. In private equity, interactions between LPs and GPs materialize in management mandates that define, from fundraising onward, an important part of how the vehicles will operate thereafter. These mandates lock in three interdependent organizational characteristics: the **value-creation objectives** pursued (financial, social, environmental), the **resources** mobilized to achieve them (teams, processes, tools), and the **incentive mechanisms (carried interest) and control mechanisms (specific committee)** that shape the behavior of investment teams.

The central idea is therefore the following: impact is realized and measured downstream in the investment chain, but it appears to be largely constructed upstream, in the fund's organizational architecture. In other words, the ability of GPs to activate the levers of sustainable finance systematically depends in part on the way their vehicles are structured.

Research question: How does the organizational structure of funds evolve when they integrate a sustainable finance logic?

Objectives: Document how funds are actually structured, identify the main organizational configurations observed, and assess what these configurations reveal about the vehicles' actual degree of sustainable commitment, beyond compliance with frameworks or regulation.

Scope of the study & methodological elements

Scope: French funds able to invest in equity, regardless of the maturity or type of underlying companies, raised since 2017.

A few figures on data collection:

- 175 GPs contacted, 91 interviewed, 43 under NDA. → 95 funds processed, 10 still awaiting data (questionnaires or second phase), after excluding fund-of-funds and single-LP vehicles, the analysis covers 89 direct funds.
- Processed sample: 55 buyout funds, 31 VC funds, 7 infrastructure funds, 2 fund-of-funds, including 22 impact funds, 42 Article 8 funds, and 30 Article 9 funds.

→ Because participation was voluntary, the sample necessarily over-represents actors that are sensitive to sustainability issues. The sample nevertheless remains **representative of the diversity of strategies in the French market**.

Documents collected: PPMs, LPAs, ESG charters (& side letters), manually coded using **more than 120 variables** drawn from the academic literature and market frameworks (the variable dictionary is available in the "Method" section).

Coding logic (2 levels):

- **Macro variables (overall architecture)**: general characteristics of the vehicles (BO/VC/Infra, ticket sizes, etc.) and level of extra-financial structuring (formalization of a sustainable investment objective, presence of key ESG screening/support/reporting processes, presence of structuring mechanisms such as the *EF Committee* and the *EF Carried*).
- **Micro / modular variables (mechanism design)**: when the mechanisms are present, detailed coding of their modalities: (i) characterization of the sustainability objective pursued, (ii) modalities of *EF Carried* indexation, (iii) composition and effective rights of the *EF Committee*, (iv) modalities for defining/monitoring sustainability objectives at the level of portfolio companies (*IMM Process*), and (v) degree of contractualization of commitments in the **fund legal documentation**.

Analytical procedure (several phases):

- **Phase 1 — Descriptives:** description of the sample and the prevalence of variables (%, averages, dispersion).
- **Phase 2 — Typology (clustering):** beyond "variable-by-variable" statistics, identification of **fund-type configurations** (combinations of practices) through a **clustering** analysis at the level of macro variables and the different micro modules. This step enables the summarization of architectural heterogeneity and the establishment of an interpretable fund typology. The process is presented in detail in the Method section.

Macro results: three sustainability profiles

The macro analysis highlights 3 profiles of sustainable structuring, describing a continuum rather than an opposition between "sustainable funds" and "non-sustainable funds."

Table 1: Three macro clusters of sustainable architecture

Cluster	n	%
C1: No Sustainability Process	13	14.6%
C2: Sustainability Process without Governance Mechanisms	30	33.7%
C3: Sustainability Process with Governance Mechanisms	46	51.7%

We observe a broadly shared basic ESG process. The majority of funds already integrate a "generic" ESG process: 89.9% apply screening through an exclusion list and perform an ESG analysis at entry, 80.9% measure sustainability indicators at portfolio-company level, and 78.7% set sustainability objectives at portfolio-company level. On the contractual side, 88.6% have an exclusion list, 85.2% a generic ESG-risk integration clause, 65.5% commit to carrying out systematic ESG analysis at entry, and 60.2% commit to annual ESG reporting.

The signature variables of the three clusters capture their differences. **Cluster 1** groups together funds **without sustainability process nor organizational mechanisms**: **only 1 fund out of 13** mentions an EF Committee, and in a symbolic form (i.e., without formal decision rights, which does not preclude practical effects), **only 1 fund out of 13** undertakes extensive legal commitments, and all the others remain centered on minimal contractualization limited to exclusion lists, without formalized screening or support processes. **Cluster 2** corresponds to funds with a structured process-oriented approach, but without dedicated governance: **63%** fully contractualize their sustainability processes, **87%** assign objectives to portfolio companies, but **83%** have no EF Carried and **97%** no EF Committee. **Cluster 3**, by contrast, combines the main building blocks of a robust architecture: **96%** undertake contractual commitments regarding their sustainability approach, **91.3%** display an explicit sustainable objective, **78.3%** have a Sustainability Team, **80.4%** an indexed carried interest, and **60.9%** an EF Committee. Even within the most structured cluster, we observe heterogeneity in the use of the four organizational mechanisms.

→ **The market therefore broadly shares a basic ESG process, but the difference lies in the depth of the associated organizational structure.**

Micro results: 4 organizational mechanisms, 12 modalities, and major design differences

Beyond the presence of organizational mechanisms (sustainable objective, IMM process, EF Carried, EF Committee), we also analyze their design, which can vary from one fund to another.

Table 2: Summary of micro modules: observed modalities and differentiating variables

Module	Observed modalities	Differentiating variables
Sustainable investment objective	Mixed focus, non-contractual, support-oriented; Contractual social focus, no target, solutions-oriented; Contractual environmental focus, support and solutions-oriented	Type of objective, additionality strategy, contractualization of intentionality, explicit targets, impact reporting and impact analysis
EF Committee	Symbolic; Controlling	Validation of EF KPIs, targets, methodology, weighting of the EF score, presence of independent members
EF Carried	Specific Outcome Targets, Thresholds, High indexation; Global Outcome Targets, Linear; Mixed Global Targets, tiered thresholds; Generic	Nature of objectives (means vs. outcomes), level of objectives (company vs. fund), share of carried indexed, offsetting across objectives, vesting mechanism (linear or tiered, thresholds)
IMM Process	Low; Intermediate: robust measurement & targets; Comprehensive	Inclusion in transaction documentation, use of frameworks, management package indexation, validation by a committee, independent measurement, continuity at exit

The distribution of modalities reveals a new source of structural heterogeneity. For the **sustainable investment objective**, the dominant modality is the **contractual environmental focus** (**26 funds out of 52**, i.e. **50%**), ahead of the **mixed, non-contractual focus** (**14 funds**, **26.9%**) and the **contractual social focus without an explicit target** (**12 funds**, **23.1%**): the sustainable intention is therefore explicit, but not with the same level of enforcement. For the **EF Committee**, **controlling committees** dominate (**20 funds out of 30**, i.e. **60%**) over symbolic committees (**10 funds**, **40%**): when this module exists, it is therefore more often substantive.

The distribution of **EF Carried** is the most dispersed: **15 funds out of 42** (**35.7%**) fall into an EF carried-interest structure calibrated to specific outcome targets, **12** (**28.6%**) into a **generic** modality, **10** (**23.8%**) into a **global outcome-target** logic, and **5** (**11.9%**) into a **tiered** mechanism. Finally, the **IMM process** is distributed across **31 funds out of 70** (**44.3%**) in the **low** modality, **21** (**30%**) in the **comprehensive** modality, and **18** (**25.8%**) in the **intermediate** modality, confirming that the structuring of target-setting and measurement of sustainability-linked objectives remains highly uneven across funds.

→ **Two funds may display the same mechanism but with a different design: what matters is not only the presence of the mechanism, but also its precise modality and the level of coercion it implies.**

Organizational typology: 8 signatures and strong dispersion

The structuring of a vehicle relies on a combination of mechanisms. We therefore conclude with a final analysis that brings out structuring profiles based on the joint use of the modalities of these mechanisms.

Table 3: Distribution of organizational signatures by SFDR classification and impact status

Signature	N	Non-impact	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
No structured approach	14	14 (21%)	0 (0%)	2 (50%)	8 (62%)	4 (10%)	0 (0%)
Minimal IMM Process - no E&S focus	11	11 (16%)	0 (0%)	1 (25%)	3 (23%)	7 (17%)	0 (0%)
Internal Resources	18	17 (25%)	1 (5%)	1 (25%)	2 (15%)	13 (31%)	2 (7%)
Mixed focus - generic EF carried + IMM	9	8 (12%)	1 (5%)	0 (0%)	0 (0%)	8 (19%)	1 (3%)
Env. focus - symbolic structuring	7	6 (9%)	1 (5%)	0 (0%)	0 (0%)	6 (14%)	1 (3%)
Env. focus - robust incentive	8	5 (7%)	3 (14%)	0 (0%)	0 (0%)	2 (5%)	6 (20%)
Env. focus - comprehensive architecture	11	2 (3%)	9 (41%)	0 (0%)	0 (0%)	0 (0%)	11 (37%)
Social focus - comprehensive architecture	11	4 (6%)	7 (32%)	0 (0%)	0 (0%)	2 (5%)	9 (30%)
Total	89	67 (100%)	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

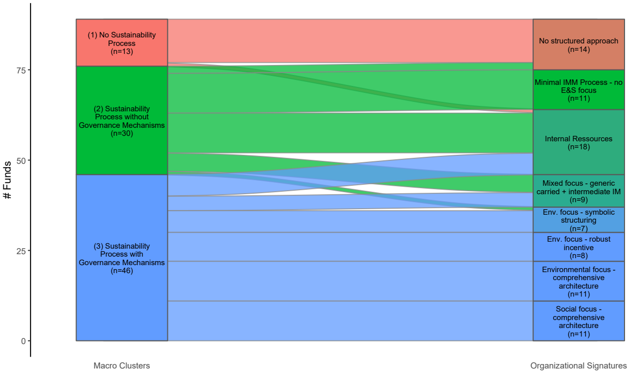
The final typology identifies eight organizational signatures, and shows strong dispersion even within the macro sets themselves: the cluster *Sustainability Process without Governance Mechanisms* is distributed across **5 signatures**, and the cluster *Sustainability Process with Governance Mechanisms* across **6 signatures**. Three broad families emerge: **weakly structured architectures** (*No structured approach*, *Minimal IMM Process - no E&S focus*: **25 funds, 28.1%**); **intermediate forms** (*Internal Resources*, *Mixed focus - generic EF carried + IMM*, *Env. focus - symbolic structuring*: **34 funds, 38.2%**); and **robust or comprehensive architectures** (*Social focus - comprehensive architecture*, *Environmental focus - comprehensive architecture*, *Env. focus - robust incentive*: **30 funds, 33.7%**).

Labels and regulation capture this heterogeneity only imperfectly. 90% of Article 9 funds and **90%** of impact funds are concentrated in the most robust architectures, but **4 Article 9 funds out of 30 (13%)** retain minimal or symbolic structuring, **4 Article 8 funds out of 42 (10%)** instead display a comprehensive architecture, and **11 non-impact funds out of 67 (17%)** already have a signature structured around a sustainable objective.

→ **SFDR categories and "impact" status provide a useful first signal, but they do not properly capture the dispersion of the organizational architectures observed.**

Organizational typology: visual representation

The following figure enables an analysis of how the initial macro-level decomposition is distributed within the final organizational typology. As a reminder, we had formed 3 clusters from the surface-level variables presented in the funds' commercial documentation: no sustainability process, sustainability process without associated governance, and sustainability process with associated governance.



Reading note: the color of the cells in the organizational typology corresponds to a weighted blend of the colors of the macro clusters composing them, as a function of the relative share of each macro cluster in the signature considered.

Comment. Logically, almost all funds in macro cluster 1 (no sustainability process) display the organizational signature "No structured approach," while those in macro cluster 3 (sustainability with governance) display one of the more comprehensive signatures.

It is particularly interesting to note (i) substantial organizational heterogeneity within macro clusters 2 and 3, which are distributed across 5 and 6 organizational signatures respectively, and (ii) the fact that the types "Internal Resources," "Mixed focus - generic EF carried + IMM," and "Env. focus - symbolic structuring" are shared across these two clusters. **This reinforces the idea of significant organizational heterogeneity behind the display of a structured sustainability process and the use of certain governance mechanisms.**

Recommendations, implications, and perspectives

Conclusion. The first finding of the study is the existence of a shared basic ESG process, widely used across all funds. The second, and main, result is the heterogeneity of the organizational structuring observed around sustainability issues across funds, summarized in the typology presented above. Some funds displaying a comprehensive or intermediate structuring, with precise and sometimes finely calibrated mechanisms that reflect genuine sensitivity to sustainability issues.

Implications. The rapid development of these sustainable investment strategies and the growing willingness to broaden the investor base of these funds, particularly toward retail investors, reinforce the need to **better understand how vehicles are structured** more generally. Sustainability adds a layer of complexity to an asset class already perceived as opaque by non-specialists: it therefore becomes useful to have a **common analytical framework** for funds, based on the vehicle's **sustainability objectives**, the **dedicated resources**, the **incentive and control mechanisms used and why**, and also on the **precise design** of these mechanisms and their **internal logic** within the fund architecture.

For regulators, industry associations, LPs, and GPs. The study highlights a useful blind spot to be filled: the **organizational structuring** of funds. In France, much industry-level work has been carried out to support the development of sustainable finance within this asset class, notably under the auspices of France Invest, FIR, PRI, or the Institut de la Finance Durable. However, this work focuses on the relationship between the GP and portfolio companies. PRI reporting, for instance, mostly focuses on GP practices. SFDR regulation, for its part, mostly focuses on outcomes and disclosure. A clear and common disclosure layer is still missing on this organizational dimension, even though the heterogeneity observed often appears in *detailed clauses* that are not easily accessible to the public. For LPs and GPs, the SFDR signal or an "impact" positioning is therefore not sufficient to summarize a fund's actual structuring: the mechanisms must be read at the level of their **design** and their **combination**. Establishing such a reading grid would enable the demonstration of industry actors' commitment and increase their legitimacy with investors and stakeholders.

Limitations. The sample remains limited in size: it is sufficient to bring out regularities and a useful typology, but it does not claim to represent the market as a whole. At this stage, the dataset also does not enable the direct assessment of the effect of the architectures observed on the vehicles' financial or extra-financial performance.

Extensions. This database opens up several natural next steps: analysis of team composition, incorporation of fundraising date to document the historical dynamic, **matching** with ESG and financial performance data, with actual portfolio composition, with disclosure and communication data, and then a progressive extension of the framework at an international scale.

Methodology

Data collection & analysis

Scope: French funds able to invest in equity, regardless of the maturity or type of underlying companies, raised since 2017.

A few figures on data collection:

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Processing frameworks: a functional classification

The frameworks cited by funds are too heterogeneous to be compared directly as they stand. The approach therefore does not consist in comparing funds by *framework name*, but by the **functions** served by these frameworks. Concretely, each framework is associated with one or several analytical functions, and each fund is then **coded as covering** the functions corresponding to the frameworks it mobilizes. A fund is thus considered to have a given function as soon as at least one of the observed frameworks is associated with it. This operation enables the translation of heterogeneous references into a common comparison space and the production of a harmonized *fund-by-function* representation that is more robust and more interpretable for cross-fund analysis.

Code	Function	Definition
CMT	Market initiative / collective commitment	Participation in a collective initiative, charter, or coalition structuring the market.
GOAL	Goal-setting / alignment framework	Framework used to formulate, frame, or align sustainability or transition objectives.
CLAS	Classification / label / eligibility	Tool used to label, classify, or qualify assets, funds, or activities.
DISC	Disclosure / reporting	Framework for transparency, disclosure, or external reporting.
IMM	Impact management framework	Doctrine, principles, or process structuring impact management in investment.
MET	Metrics / measurement / accounting	Tool used to measure, quantify, or account for externalities, emissions, or impact.
CLIM	Climate / transition	Framework centered on climate, decarbonization, alignment, or transition.
POL	Science-policy / knowledge base	Framework producing a public or scientific knowledge, expertise, or framing base.
REG	Regulatory framework / public instrument	Public, quasi-regulatory framework, or one directly anchored in a regulatory architecture.

The list of frameworks observed in the study is available in the appendix.

Variable definitions (1/5) — general & macro

Table 4: Variable dictionary — General & Macro

Variable	Module	Source doc	Definition
Flag BO	General	Legal	Buyout fund flag. Criteria = profitable / mature targets
Flag Infra	General	Legal	Infrastructure fund flag
Flag Majo	General	Legal	Fund mainly takes majority positions (>50% of commitments)
Flag Mino	General	Legal	Fund mainly takes minority positions (>50% of commitments)
Flag VC	General	Legal	Venture capital fund flag. Criteria: unprofitable / young targets
Flag Impact	General	Commercial	Fund presented as an impact fund
Ticket Max	General	Legal	Maximum ticket size (EUR m)
Ticket Min	General	Legal	Minimum ticket size (EUR m)
EF Committee	Macro	Legal	External governance body in the form of an impact committee or equivalent
Sustainability Team	Macro	Commercial	Dedicated sustainability team
EF Carried Flag	Macro	Legal	The fund has a carried interest linked to EF objectives
Sustainable Investment Objective Flag	Macro	Commercial	Fund aims to contribute to environmental or social objectives
Focused	Macro	Legal	Fund is focused on a specific sector or theme
IMM: Objectives	Macro	Commercial	Flag for the setting of EF objectives / EF business plan at portfolio-company level
IMM: Measurement Process	Macro	Commercial	Flag for annual measurement process
IMM: Screening	Macro	Commercial	Flag for sustainability indicators (ESG and/or impact) integrated into investment analysis
Aspirational ESG Statement	Macro	Commercial	Aspirational ESG statement
ESG Advisory Support	Macro	Commercial	Mention of the use of ESG advisory support
Functions Covered	Macro	Commercial	Number of functions covered through frameworks
Financial Objective	Macro	Commercial	Fund aims to deliver market-rate financial returns
ESG Support Process	Macro	Commercial	Mention of an ESG support process in the PPM
ESG Screening Process	Macro	Commercial	Mention of a screening process in the PPM

Comment. All binary variables (for example, "Flag..." or "Sustainability Team") are coded 1 if the element is true/present and 0 if the element is false/absent.

Variable definitions (2/5) — EF Carried module

Table 5: Variable dictionary — EF Carried module

Variable	Module	Source doc	Definition
EF Carried amount	EF Carried	Legal	Share of carried interest indexed to sustainability criteria (% of carried interest)
Global Carried objectives	EF Carried	Legal	EF objectives are the same for all portfolio companies and measured at fund level
Outcome-based Carried objectives	EF Carried	Legal	EF objectives are outcome-based objectives
Weighting by amount invested	EF Carried	Legal	Portfolio-level weighting based on investment ticket sizes
Offsetting across objectives	EF Carried	Legal	Possible offsetting across lines or objectives
Linear EF Carried	EF Carried	Legal	Linear vesting between trigger and full-vesting thresholds
EF Carried trigger score	EF Carried	Legal	Threshold from which extra-financial carried interest starts being paid (% EF score)
EF Carried paid at trigger	EF Carried	Legal	Share of EF Carried paid at the trigger threshold (% of EF Carried)
Intermediate vesting tier	EF Carried	Legal	Share of EF Carried paid at the intermediate threshold (% of EF Carried)
Full vesting cap for EF Carried	EF Carried	Legal	Threshold from which EF Carried is fully vested (% EF score)

Comment. All binary variables (for example, "Flag..." or "Sustainability Team") are coded 1 if the element is true/present and 0 if the element is false/absent. These variables are coded for funds with carried interest indexed to sustainability objectives (EF Carried module).

Variable definitions (3/5) — EF Committee module

Table 6: Variable dictionary — EF Committee module

Variable	Module	Source doc	Definition
Advisory EF Committee	EF Committee	Legal	Committee decisions are not binding on the management company
Decision: Change in method & KPIs	EF Committee	Legal	EF Committee validates any change in method, KPIs, or targets
Decision: EF KPI targets	EF Committee	Legal	EF Committee validates the targets to be reached for EF KPIs
Decision: EF Carried payout	EF Committee	Legal	EF Committee validates the payout of EF Carried
Decision: Investment eligibility	EF Committee	Legal	EF Committee validates the eligibility of the investment for the fund's EF thesis
Decision: EF KPIs monitored	EF Committee	Legal	EF Committee validates the EF KPIs monitored for each portfolio company
Decision: Beneficiary organizations	EF Committee	Legal	EF Committee validates the choice of beneficiary if EF Carried is not vested
Decision: Impact orientation	EF Committee	Legal	EF Committee gives a simple opinion on the fund's extra-financial orientation
Decision: OTI measurement	EF Committee	Legal	EF Committee validates the independent third party carrying out EF reporting
Decision: Weighting of impact score	EF Committee	Legal	EF Committee validates KPI weighting / the definition of the EF score
Decision: Validation of measurement	EF Committee	Legal	EF Committee reviews and validates the final achievement of EF objectives
Committee: Independent members	EF Committee	Legal	EF Committee includes independent members
Committee: LPs	EF Committee	Legal	LPs represented on the EF Committee

Comment. All binary variables (for example, "Flag..." or "Sustainability Team") are coded 1 if the element is true/present and 0 if the element is false/absent. These variables are coded for funds with an EF Committee.

Variable definitions (4/5) — Sustainable objective module

Table 7: Variable dictionary — Sustainable objective module

Variable	Module	Source Doc	Definition
Financial Additionality	Sust. Objective	Commercial	Fund primarily characterized by financial additionality
Operational Additionality	Sust. Objective	Commercial	Fund primarily characterized by extra-financial additionality
Impact Analysis	Sust. Objective	Legal	Obligation to conduct an analysis of impact criteria before each transaction
Has Any Target	Sust. Objective	Legal	Fund has a global extra-financial objective backed by a market framework
Impact Intentionality	Sust. Objective	Legal	Mention of the intention to generate positive impact in the investment policy
Environmental Objective	Sust. Objective	Commercial	Fund has an environmental objective
Social Objective	Sust. Objective	Commercial	Fund has a social objective
Process IMM	Sust. Objective	Legal	IMM process reflected in the fund's legal documentation (not only in the SFDR annex)
Impact Reporting	Sust. Objective	Legal	Obligation to conduct periodic impact reporting

Comment. All binary variables (for example, "Flag..." or "Sustainability Team") are coded 1 if the element is true/present and 0 if the element is false/absent. These variables are coded for funds indicating that they pursue a sustainable objective (Sustainable Objective module).

Variable definitions (5/5) — Contractualization module & IMM module

Table 8: Variable dictionary — Sustainable objective module & transversal module

Variable	Module	Source Doc	Definition
ESG Analysis	Transverse	Legal	Obligation to conduct a general ESG analysis before each investment.
ESG DD	Transverse	Legal	Reference to external ESG due diligence (1 if yes, 0 if no, 0.5 if case by case)
ESG in Shareholder's Agreement	Transverse	Legal	Mention of ESG criteria being included in shareholder agreements (1 if yes)
Exclusions	Transverse	Legal	Fund has a classic sectoral exclusions list (tobacco, oil, illegal activity, alcohol, gambling, pornography, etc.)
ESG Integration	Transverse	Legal	Mention of ESG criteria being integrated into investment analysis
Systematic ESG Process	Transverse	Legal	Mention of an ESG process to be implemented systematically (1 if yes)
ESG Reporting	Transverse	Legal	Obligation to conduct periodic reporting on extra-financial criteria
Share Contractual	Transverse	Legal	Share of frameworks included in the fund's legal documentation (%)
IMM: Doc Transaction	IMM	Commercial	Flag for systematic integration of sustainability objectives in companies' contractual documentation (0.5 = scientific advisor, no framework)
IMM: Exit Management	IMM	Commercial	Flag for systematic consideration of continuity of the sustainability strategy in the exit strategy
IMM: ManPack	IMM	Commercial	Flag for systematic integration of sustainability objectives into managers' incentive mechanism
IMM: Independent Measurement	IMM	Commercial	Flag for measurement performed by an independent party
IMM: Frameworks	IMM	Commercial	Systematic use, where applicable, of a framework to define objectives (0.5 = scientific advisor but no framework)
IMM: EF Committee Validation	IMM	Commercial	Flag for systematic validation of objectives and measurement method by an EF Committee

Comment. All binary variables (for example, "Flag..." or "Sustainability Team") are coded 1 if the element is true/present and 0 if the element is false/absent. For the IMM module, these variables are coded when the fund systematically sets sustainability objectives at the level of its portfolio companies (IMM module).

Cluster analysis: principles and application to our fund database

General objective. Cluster analysis enables the identification of **types of funds** displaying **similar organizational architectures with respect to sustainability issues** (objectives, resources, incentive and control mechanisms), based on the variables observed in the documents.

Why not limit the analysis to individual variables? A proportions analysis (e.g. "x% of funds have an EF Committee") describes **isolated practices**, but it does not say how these practices **combine** into coherent configurations. In reality, sustainability mechanisms often operate as a **system**: some building blocks reinforce one another (complementarities), others substitute for one another, and particular combinations may signal more or less developed approaches. Cluster analysis synthesizes this complexity by bringing out **complete** and comparable profiles.

Method. This is an **exploratory** and **unsupervised** statistical method: it does not start from any predefined typology, but automatically groups together funds **that resemble each other** across the full set of variables, maximizing **within-group** homogeneity and **between-group** heterogeneity. The result is an **empirical typology** of practices, independent of ex ante categories (SFDR, labels, declarations), since it is based only on the practices observed in the documents.

We apply a two-level approach to distinguish **structuring choices** from the **modalities of mechanisms**:

- **Macro step:** group funds according to their **broad architectural choices** (e.g. existence of a sustainable objective, presence of associated resources and governance/incentive mechanisms). This step describes the **overall degree of structuring** of the sustainability approach.
- **Micro step:** within the "committed" funds, characterize **how** the mechanisms are actually designed and operationalized: degree of contractual integration, sophistication of processes (IMM), design of incentives (EF carried), effective role of governance (committee), etc. This step brings out **configuration-types** (e.g. substantive vs symbolic mechanisms, demanding vs generic designs).
- **Hierarchical reading:** micro choices are interpreted **conditionally** on macro choices (one does not expect the same level of micro sophistication in a fund without a general sustainability base).

Deliverable: a simple mapping of **fund types** and **dominant designs**, then used to (i) compare practices, (ii) identify the most frequent and most distinctive combinations, and (iii) prepare phase-2 performance analyses.

Description

Sample description — General

Variable	Source Doc	n	n = 1	Average
Flag BO	Legal	89	53	59,6%
Flag Infra	Legal	89	7	7,9%
Flag Majo	Legal	89	40	44,9%
Flag Mino	Legal	88	47	53,4%
Flag VC	Legal	89	29	32,6%
Ticket Max	Legal	74	–	19,601
Ticket Min	Legal	75	–	5,078
Impact	Commercial	89	22	24,7%
SFDR: Art. 8	Legal	89	42	47.2%
SFDR: Art. 9	Legal	89	30	33.7%

Comment. The sample used for the analysis therefore consists of 53 buyout funds, 29 VC funds, and 7 infrastructure funds or funds with infrastructure-type underlying assets. We excluded single-LP funds, as well as corporate funds, in order to focus the analysis on funds managed by independent management companies. The distribution between funds taking majority and minority positions is fairly balanced.

Sample description — Macro

Variable	Source Doc	n	n = 1	Average
EF Committee	Legal	89	30	33,7%
Sustainability Team	Commercial	89	48	53,9%
Flag EF Carried	Legal	89	42	47,2%
Flag Sustainable Investment Objective	–	89	52	58,4%
Focused	–	89	43	48,3%
IMM: Objectives	Commercial	89	70	78,7%
IMM: Measurement Process	Commercial	89	72	80,9%
IMM: Screening	Commercial	89	80	89,9%
Aspirational ESG Statement	Commercial	89	75	84,3%
ESG Advisory Support	Commercial	89	38	42,7%
Functions Covered	–	89	–	2,607
Financial Objective	Commercial	89	89	100,0%
ESG Support Process	Commercial	89	45	50,6%
ESG Screening Process	Commercial	89	58	65,2%

Comment. The macro part of the analysis characterizes the overall structuring of the funds in the sample and their sensitivity to sustainability issues (contribution to a sustainable objective, the most important stages of the IMM process, etc.), without going into the detail of the degree of contractualization or the functioning of the mechanisms (additionality strategy, modalities of the EF Committee and EF Carried, degree of ambition of the objectives, etc.).

All funds in the sample target market-rate financial returns. The majority have a common process base around sustainability issues: screening at entry, setting EF objectives, and annual measurement. The organizational structuring around this process is more heterogeneous: 33.7% have an EF Committee, 47.2% have carried interest indexed to an EF score, and 58.4% explicitly state that they intend to contribute to an environmental or social objective through their investment activities.

From the perspective of a potential investor, these variables represent a quickly grasp of how the vehicle is structured around sustainability issues, often as presented in the fund's commercial documentation.

Sample description — EF Carried module

Variable	Source doc	n	n = 1	Mean	Std. dev.
Linear EF Carried	Legal	42	32	76.2%	–
Global Carried objectives	Legal	42	14	33.3%	–
Outcome-based Carried objectives	Legal	42	26	61.9%	–
Offsetting across objectives	Legal	42	18	42.9%	–
Weighting by amount invested	Legal	42	24	57.1%	–
EF Carried amount	Legal	42	–	33.4%	0.21
EF Carried trigger score	Legal	42	–	30.1%	0.30
EF Carried paid at trigger	Legal	42	–	20.7%	0.30
Intermediate vesting tier	Legal	8	–	81.3%	0.22
Full vesting cap for EF Carried	Legal	42	–	91.0%	0.11

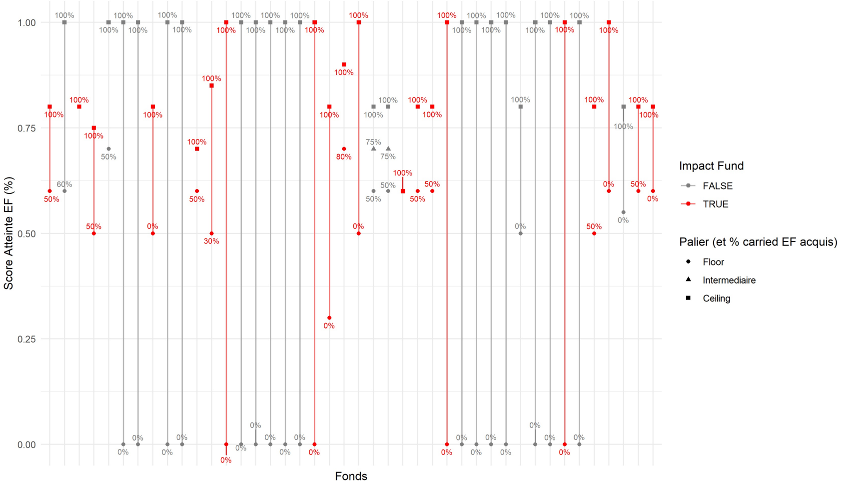
Comment. The variables in the "EF Carried" module concern the 42 funds that have a carried interest indexed to sustainability objectives and enable the analysis of the modalities of that indexation. In our sample, the average amount of indexed carried interest (or sometimes called *"at risk"*) is 33.4% (with a large standard deviation of 21 points).

We observe substantial heterogeneity in the indexation modalities: for most funds, carried interest is indexed to a score reflecting achievement of sustainability objectives defined specifically for each portfolio company (62%), weighted by investment ticket sizes (57.1%) or by a simple line-by-line average (42.9%).

We also observe substantial heterogeneity in the payout modalities of carried interest: on average, EF Carried starts being paid at an EF score of 30.1%, and the whole of EF Carried is on average fully vested from a score of 91%, often with linear variation between the thresholds (76.2%). Eight funds operate with tiers, with an intermediate tier on average around 81.3% achievement of the EF score. The standard deviations on these values are large (0.11 to 0.30), indicating substantial heterogeneity across funds, particularly on the trigger threshold and the amount paid at that threshold.

The next slide provides a visual representation of these modalities, and clustering will enable a finer analysis of the co-occurrence patterns among these different variables in order to bring out the "typical configurations" of EF Carried.

Carried module — Threshold structure



Comment. This figure represents the payout mechanics of carried for the funds in the sample. On the horizontal axis, each vertical line represents a fund whose carried interest is indexed to EF objectives, with the trigger (*floor*) and full-vesting (*ceiling*) thresholds expressed as a function of achievement of the fund's EF score. Next to each threshold, the % of EF Carried vested when that threshold is reached is indicated. A continuous line indicates linear variation between thresholds.

Sample description — EF Committee module

Variable	Source doc	n	n = 1	Mean
Advisory EF Committee	Legal	30	29	96.7%
Decision: Change in method & KPIs	Legal	30	11	36.7%
Decision: EF KPI targets	Legal	30	20	66.7%
Decision: EF Carried payout	Legal	30	4	13.3%
Decision: Investment eligibility	Legal	30	8	26.7%
Decision: EF KPIs monitored	Legal	30	20	66.7%
Decision: Beneficiary organizations	Legal	30	6	20.0%
Decision: EF orientation	Legal	30	4	13.3%
Decision: OTI measurement	Legal	30	3	10.0%
Decision: Weighting of EF score	Legal	30	15	50.0%
Decision: Validation of measurement	Legal	30	13	43.3%
Committee: Independent members	Legal	30	19	63.3%
Committee: LPs	Legal	30	18	60.0%

Comment. The variables in the "EF Committee" module concern the 30 funds with an impact committee or equivalent. In almost all cases (29 funds out of 30), this committee is advisory and non-binding. Beyond that, we observe substantial heterogeneity in the responsibilities assigned to these committees: the most common is the validation of sustainability KPIs monitored at portfolio-company level (20 funds), the associated objectives (20 funds), and their annual or measurement at liquidation (15 funds). We also observe substantial heterogeneity in the composition of these committees. 60% of them grant voting seats to LPs, and 63.3% to independent members. Clustering will enable a finer analysis of the co-occurrences among these different variables in order to bring out the "typical configurations" of these committees.

Sample description — IMM Process module

Variable	Source doc	n	n = 1	Mean
IMM: Frameworks	Commercial	70	11	19.3%
IMM: EF Committee Validation	Commercial	70	21	30.0%
IMM: Doc Transaction	Commercial	70	28	52.1%
IMM: ManPack	Commercial	70	12	18.6%
IMM: Independent Measurement	Commercial	70	36	51.4%
IMM: Exit Management	Commercial	70	8	11.4%

Comment. The variables in the "IMM Objectives" module concern the 70 funds that define and assign sustainability objectives to their portfolio companies. We observe substantial heterogeneity in the processes used to define and monitor these sustainability objectives. In particular, a majority of funds adopt independent measurement of these objectives (36) and include them in the contractual documentation of portfolio companies (28 systematically, and roughly ten additional cases in an aspirational or occasional way). A minority systematically use frameworks to ensure the ambition of the objectives defined with portfolio companies (11), explicitly integrate them into managers' management packages (12), and take the continuity of the strategy into account at the time of disposal of the portfolio company (8). Clustering will enable a finer analysis of the co-occurrences among these different variables in order to bring out the "typical configurations" of these sustainability-objective management processes.

Sample description — Sustainable Investment Objective module

Variable	Source doc	n	n = 1	Mean
Financial Additionality	Commercial	52	35	67.3%
Operational Additionality	Commercial	52	20	38.4%
Impact Analysis	Legal	50	26	52.0%
Has Any Target	–	52	5	9.6%
Impact intentionality	Legal	50	32	64.0%
Environmental objective	Commercial	52	46	88.5%
Social objective	Commercial	52	15	28.8%
IMM Process	Legal	50	31	62.0%
Impact reporting	Legal	50	23	46.0%

Comment. Among the 52 funds indicating that they contribute to a sustainable objective (environmental or social) through their investment activities, 35 have a financial additionality strategy (financing solutions that contribute directly to the objectives) and 20 have an operational additionality strategy (supporting portfolio companies in reducing their negative footprint or transforming practices through a sustainable transformation strategy). Three funds have a mixed strategy. Most funds pursue an environmental objective (46), and a significant share pursue a social objective (15). The majority of funds formalize their intention to contribute to a sustainable objective (intentionality) in their contractual documentation (32, 64%) and integrate specific clauses detailing their Impact Management & Measurement (IMM) process beyond the SFDR annex (31, 62%). Funds with a broad environmental or social objective backed by a scientific and/or industrial framework remain a minority (5 vehicles). Clustering will enable a finer analysis of the co-occurrences among these different variables in order to bring out the "typical configurations" of the formal contractual commitments linked to the sustainability objectives pursued by the funds.

Sample description — ESG Contractualization module

Variable	Source doc	n	n = 1	Mean	Std. dev.
ESG Analysis	Legal	87	57	65.5%	–
ESG DD	Legal	87	24	31.0%	–
ESG in SHA	Legal	87	17	19.5%	–
Exclusions	Legal	88	78	88.6%	–
ESG Integration	Legal	88	75	85.2%	–
Systematic ESG Process	Legal	87	43	49.4%	–
ESG Reporting	Legal	88	53	60.2%	–
Share Contractual	–	89	–	26.3%	0.37

Comment. These variables concern the 89 funds in the sample. They capture contractual commitments undertaken in the area of sustainable investment, whether or not the fund seeks to contribute to a sustainable objective. All of them come from the vehicles' legal documentation.

The vast majority of funds apply an exclusion list (78, 88.6%) and integrate an ESG risk-integration clause, often generic (75, 85.2%). Most funds commit to carrying out a specific ESG analysis for each transaction and detail this analytical process in their contractual documentation (57, 65.5%), and to carrying out specific ESG reporting (60.2%). Within this analysis, 24 funds (31%) explicitly commit to carrying out external ESG DD for each transaction.

Funds integrating clauses linked to their support process (ESG in the SHA, systematic onboarding of portfolio companies through the ESG process) represent a minority share (17 and 43 respectively) but a significant one.

Clustering will enable a finer analysis of the co-occurrences among these different variables in order to bring out the "typical configurations" of contractualization concerning sustainability issues.

Macro Structure

Macro clusters: sustainability commitments

The analysis produces a partition of the sample into 3 groups of funds.

Cluster 1: No Sustainability Process

- Size: 13 funds (14.6%)
- Description: funds not integrating any sustainable-finance practice.
- Signature variables: none. All variables are zero or very significantly below the average of the other clusters.

Cluster 2: funds with a structured sustainability process, without dedicated governance

- Size: 30 (33.7%)
- Description: funds developing sustainable-finance practices (screening, support, reporting), without a sustainable objective or governance mechanism dedicated to sustainability issues.
- Signature variables: Aspirational ESG Statement, annual measurement of ESG indicators, setting objectives at portfolio-company level, systematic screening process.

Cluster 3: funds with a structured sustainability process and dedicated governance

- Size: 46 (51.7%)
- Description: funds developing sustainable-finance practices (screening, support, reporting), with a sustainable objective / dedicated governance mechanisms.
- Signature variables: signature variables of cluster 2 + Sustainability Team, carried interest indexed to sustainability objectives, EF Committee ruling on certain decisions related to extra-financial performance.

Macro — Cluster 2: signature variables (over-represented)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Variable	**score**	p_c
IMM: Measurement Process	0.455	93.3%
IMM: Objectives	0.388	86.7%
IMM: Screening	0.346	100.0%
ESG Screening Process	0.209	63.3%
Aspirational ESG Statement	0.179	83.3%
ESG Support Process	-0.003	36.7%
ESG Advisory Support	-0.015	30.0%
Sustainability Team	-0.063	36.7%
Functions Covered	-0.091	-0.364
Sustainable Investment Objective Flag	-0.123	33.3%

Comment. Cluster 2 is distinguished by a high **IMM structuring** (screening, objectives, measurement: $p_c \in [0.87; 1.00]$), while the organizational and contractual building blocks appear **weaker or non-discriminating** (Sustainability Team, support, and advisory support close to the average of the other clusters). The profile suggests an approach mainly **equipped at the level of operational processes**, without formal intentionality and without associated governance mechanisms (EF Carried, EF Committee).

Macro — Cluster 3: signature variables (over-represented)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Variable	score	p_c
Functions Covered	1.276	54.7%
Sustainable Investment Objective Flag	0.746	91.3%
Flag EF Carried	0.721	80.4%
Sustainability Team	0.561	78.3%
ESG Support Process	0.556	73.9%
EF Committee	0.554	60.9%
ESG Screening Process	0.531	84.8%
IMM: Objectives	0.523	95.7%
IMM: Measurement Process	0.490	95.7%
ESG Advisory Support	0.480	63.0%

Comment. Cluster 3 combines a **high and multidimensional structuring**. At the resource level, these funds exhibit a systematic IMM base (screening, measurement, objectives) ($p_c \in [0.96; 1.00]$), as well as a substantial level of detail on ESG screening and support operational processes in the marketing documentation ($p_c = 84.8\%$ and $p_c = 73.9\%$). Most of these funds have an internal Sustainability Team ($p_c = 78.3\%$, $+56.1$ points relative to the average of the other clusters) and mention relying on sustainability experts in their investment and support processes.

We also observe more formalized governance on sustainability issues for these funds, with the explicit mention of contributing to a sustainable objective ($p_c = 91.3\%$, $+74.6$ points relative to the average of the other clusters), a formal incentive mechanism through indexation of part of their carried interest to sustainability indicators ($p_c = 80.4\%$, $+72.1$ points relative to the average of the other clusters), and an EF Committee or equivalent ($p_c = 60.9\%$, $+55.4$ points relative to the average of the other clusters).

MCA: building a map of funds and clusters

Why use MCA dimensionality reduction? Our variables are mostly **categorical** (often binary: 0/1). MCA (Multiple Correspondence Analysis) is a descriptive method that enables the **summarization of this information in a 2-dimensional map**, in order to visualize **proximities** and **oppositions** between funds, and thus to visualize the distribution of clusters graphically.

Methodological principle. MCA transforms each variable into **two modalities** (e.g. "IMM: Screening = 1" vs "= 0") and then seeks the **axes** that best separate fund profiles (in terms of distances between configurations of modalities). The first two axes are those that **explain the greatest share of variation** in the data. In our case, they explain 56.6% of the variation in the data. **This is therefore only an imperfect visual representation.**

What does the map show?

- **Each point** represents a fund.
- Two nearby funds have **similar practices** (the same observed modalities).
- Two distant funds have **different architectures**.
- Funds located in the direction of a modality tend to **carry that modality more strongly**.

How should the axes be interpreted? The axes have no a priori label: they are interpreted from the **modalities that contribute most** to their construction (see the following slides). In our case, axis 1 mainly reflects a **gradient of operational structuring** (IMM base and ESG processes), while axis 2 more strongly reflects **organizational formalization / governance** (EF Committee, focus, sustainable investment objective, etc.).

MCA — Construction of axes: "Operational Sustainability Structuring"

Reading grid. MCA axes are linear combinations maximizing the explained inertia (or the distance between points). The **Contrib** column indicates the relative contribution of each modality to the dimension (the strongest contributions structure the axis). The **Coord1** and **Coord2** columns correspond to the factor coordinates of the modalities on the first two dimensions (sign = side of the axis, amplitude = distance from the center).

Table 9: Modalities contributing most to Dimension 1 - Horizontal Axis (MCA)

Modality	Contrib	Coord Dim 1	Coord Dim 2
IMM: Objectives = 0	11.235	-1.615	0.388
IMM: Measurement Process = 0	10.668	-1.663	0.032
IMM: Screening = 0	9.288	-2.133	0.396
Aspirational ESG Statement = 0	8.500	-1.636	0.726
ESG Screening Process = 0	6.615	-0.970	0.492
Sustainable Investment Objective Flag = 0	4.645	-0.744	-0.708
Flag EF Carried = 1	4.639	0.698	0.395
ESG Support Process = 0	4.265	-0.654	0.274
ESG Support Process = 1	4.170	0.639	-0.268
Flag EF Carried = 0	4.145	-0.624	-0.353
ESG Advisory Support = 1	3.991	0.681	-0.088
ESG Screening Process = 1	3.536	0.518	-0.263
Sustainable Investment Objective Flag = 1	3.305	0.529	0.504
IMM: Objectives = 1	3.050	0.438	-0.105
Sustainability Team = 0	3.014	-0.569	0.090

Comment. Axis 1 is mainly driven by the modalities linked to the **IMM base** (IMM: Objectives/Measurement Process/Screening), as well as by the existence (or absence) of operational ESG processes. Accordingly, funds to the left of the center are funds without an operational sustainability process, while those to the right are those with the most structured process.

MCA — Construction of axes: "Sustainable Governance Formalization"

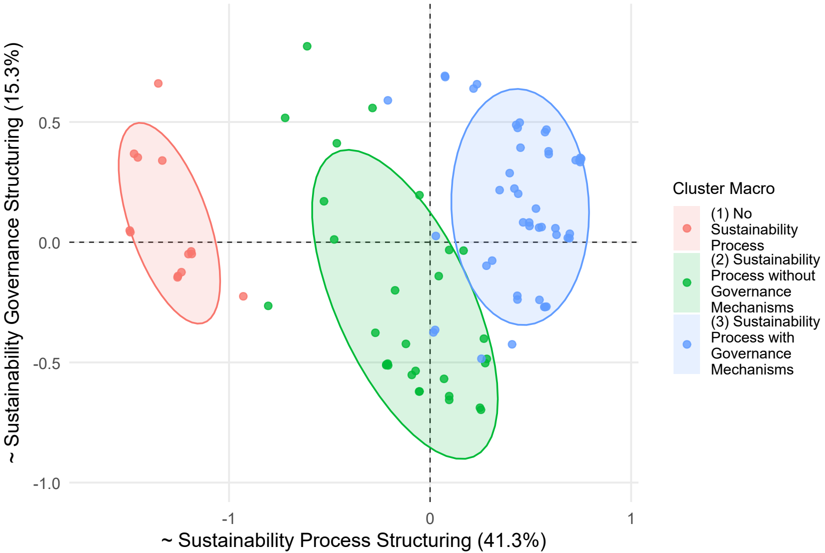
Reading grid. Same logic. MCA axes are linear combinations maximizing the explained inertia (or the distance between points). The **Contrib** column indicates the relative contribution of each modality to the dimension (the strongest contributions structure the axis). The **Coord1** and **Coord2** columns correspond to the factor coordinates of the modalities on the first two dimensions (sign = side of the axis, amplitude = distance from the center).

Table 10: Modalities contributing most to Dimension 2 - Vertical Axis (MCA)

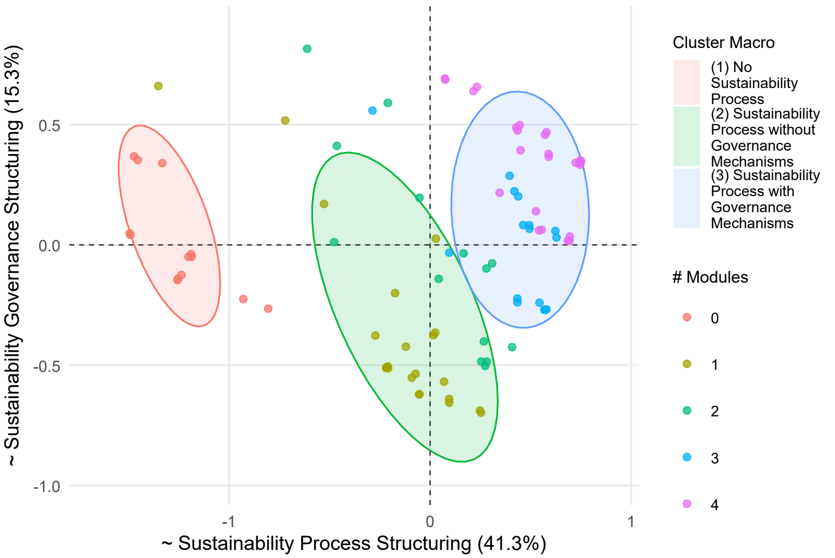
Modality	Contrib	Coord Dim 1	Coord Dim 2
EF Committee = 1	15.439	0.604	0.917
Flag Focused = 1	15.191	0.162	0.760
Flag Focused = 0	14.200	-0.151	-0.710
Sustainable Investment Objective Flag = 0	11.358	-0.744	-0.708
Sustainable Investment Objective Flag = 1	8.082	0.529	0.504
EF Committee = 0	7.850	-0.307	-0.466
ESG Screening Process = 0	4.586	-0.970	0.492
Aspirational ESG Statement = 0	4.515	-1.636	0.726
Flag EF Carried = 1	4.011	0.698	0.395
Flag EF Carried = 0	3.585	-0.624	-0.353
ESG Screening Process = 1	2.451	0.518	-0.263
ESG Support Process = 0	2.021	-0.654	0.274
ESG Support Process = 1	1.977	0.639	-0.268
IMM: Objectives = 0	1.754	-1.615	0.388
IMM: Screening = 0	0.862	-2.133	0.396

Comment. For axis 2, the strongest contributions mainly concern the **EF Committee** and **Flag Focused**, followed by the intention to contribute to a **sustainable objective**. In practice, this axis 2 captures less of the "IMM base" (which mainly structures axis 1) than **dimensions of organizational architecture and governance**: dedicated governance (committee), degree of sectoral / thematic focus (focused), and to a lesser extent the presence of EF Carried, associated with the formalization of a sustainable objective.

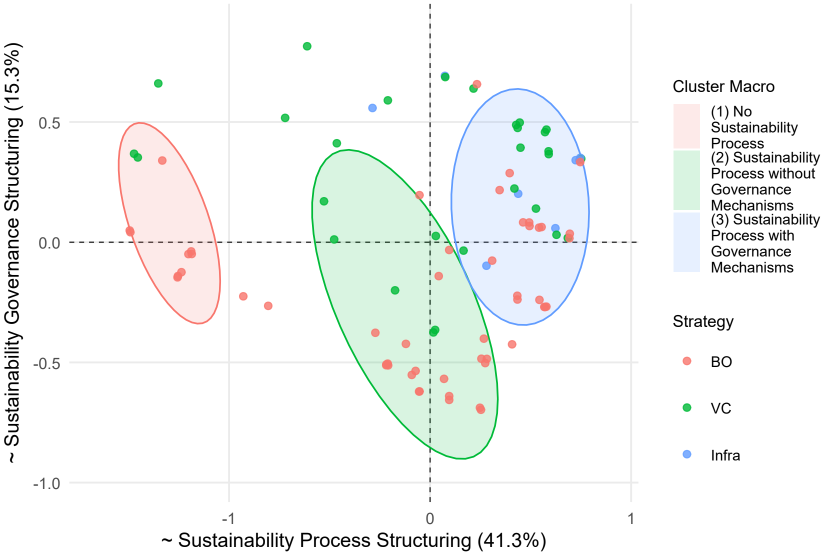
Macro — Graphical representation (MCA)



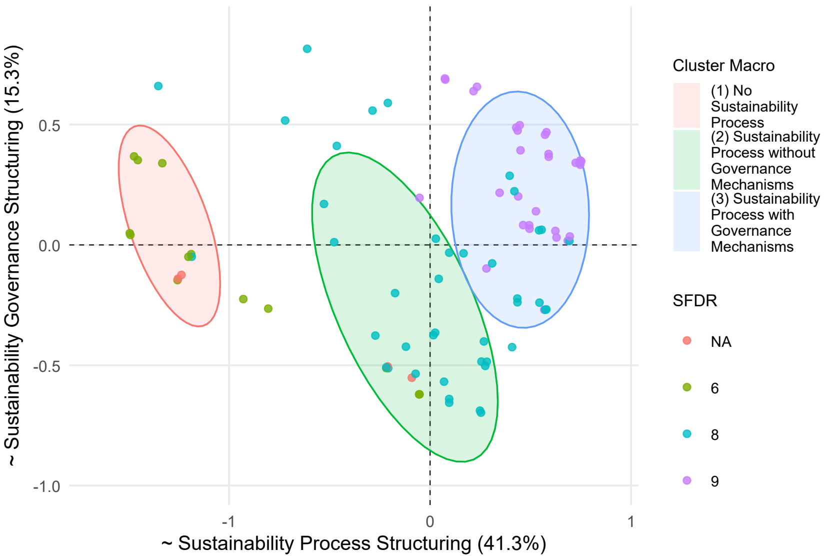
Macro — Graphical representation (MCA) — equipment



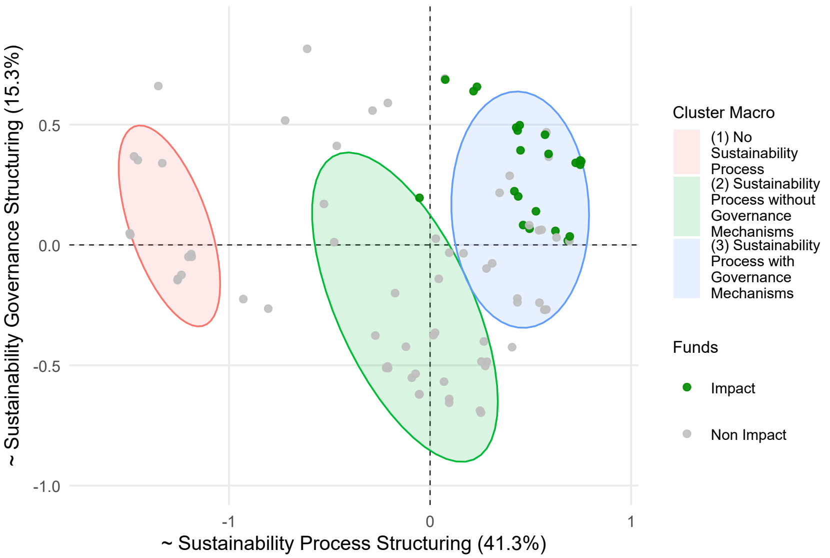
Macro — Graphical representation (MCA) — strategy



Macro — Graphical representation (MCA) — SFDR



Macro — Graphical representation (MCA) — impact



Mechanisms

Organizational mechanisms: analytical steps

This section presents the main **organizational mechanisms** (also called "modules") observed in the organizational architecture of funds: sustainable investment objective, EF Committee, EF Carried, and IMM process. We also provide an analysis of "ESG contractualization," which cuts across all funds and enables the assessment of the extent to which sustainability commitments are formalized in the vehicles' legal documentation.

Each module is studied in two stages: first through clustering on its own variables in order to bring out the main design modalities of the mechanism, and then by describing how the funds in the sample use the mechanism according to the sustainable objective pursued, their additionality strategy, and their SFDR and impact classification.

Objective: distinguish the mere presence of a mechanism from its **effective design**, and then understand how the mechanisms articulate within the overall architecture of the vehicles.

Module: ESG contractualization

Transversal module: ESG contractualization

The analysis produces a partition of the sample into 3 groups of funds.

Cluster 1: Exclusion-list only

- Size: 21 (23.6%)
- Description: funds with a mainly non-contractual ESG approach, relying on declarative or operational commitments outside the fund legal documentation. No more than an exclusion list.

Cluster 2: Contractualized sustainability process (analysis, engagement, reporting)

- Size: 54 (60.7%)
- Description: funds with a sustainability process that is largely formalized in the fund legal documentation.
- Signature variables:
  - ESG analysis embedded in the fund legal documentation,
  - ESG risk integration embedded in the fund legal documentation (aspirational),
  - Contractual ESG reporting,
  - ESG process imposed on GPs by the fund legal documentation, often with a clause in the SHA.

Cluster 3: Focus on the analysis stage

- Size: 14 (15.7%)
- Description: funds with precise clauses on the analysis component, often supported by one of the market frameworks.
- Signature variables: ESG analysis embedded in the fund legal documentation, reference to a large number of frameworks.

Transversal module: Cluster 1 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 11: Transversal — Cluster 1: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctile	global avg.
ESG in SHA	-0.163	0.0%	N/A	–	–
Exclusions	-0.165	76.2%	N/A	–	–
DD ESG	-0.338	0.0%	N/A	–	–
Share Contractual	-0.430	2.4%	moyen	58.4%	0.26
Systematic ESG Process	-0.435	4.8%	N/A	–	–
ESG Reporting	-0.438	9.5%	N/A	–	–
ESG Integration	-0.619	38.1%	N/A	–	–
ESG Analysis	-0.913	0.0%	N/A	–	–

Comment. Cluster 1 corresponds to **minimal contractualization of sustainability commitments**, based on an exclusion list (76.2% of funds, 16.5 points lower than the average of the other clusters) and sometimes on a generic ESG risk-integration clause in the investment process (38.1% of funds, 61.9 points lower than the average of the other clusters). We observe very limited ESG reporting (9.5%, 43.8 points lower than the average of the other clusters), complete absence of the other clauses, and likewise no transfer of frameworks into the fund legal documentation (very low Share Contractual).

Transversal module: Cluster 2 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 12: Transversal — Cluster 2: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctile	global avg.
ESG Reporting	0.805	92.5%	N/A	–	–
Systematic ESG Process	0.619	75.0%	N/A	–	–
DD ESG	0.354	46.2%	N/A	–	–
ESG in SHA	0.327	32.7%	N/A	–	–
ESG Analysis	0.327	82.7%	N/A	–	–
ESG Integration	0.310	100.0%	N/A	–	–
Exclusions	0.079	92.5%	N/A	–	–
Share Contractual	-0.083	25.5%	moyen	62.9%	0.26

Comment. Cluster 2 is marked by **broad contractual formalization** of sustainability commitments (reporting, ESG process deployed for each portfolio company, systematic integration of ESG risks). It evokes an **extended "standard"** contractualization across the different stages of the funds' sustainability processes.

Transversal module: Cluster 3 (signatures)

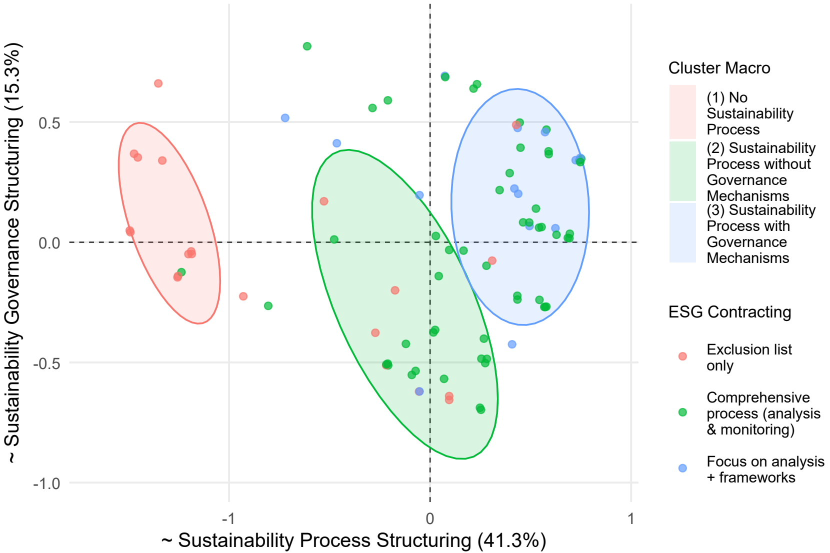
Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, $pctl$ simply indicates the share of funds with a value lower than or equal to p_c. The p_tag label provides a synthetic reading in three levels: low, medium, or high.

Table 13: Transversal — Cluster 3: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctile	global avg.
ESG Analysis	0.587	100.0%	N/A	–	–
Share Contractual	0.513	65.2%	haut	80.9%	0.26
ESG Integration	0.310	100.0%	N/A	–	–
Exclusions	0.085	92.9%	N/A	–	–
DD ESG	-0.016	21.4%	N/A	–	–
ESG in SHA	-0.163	0.0%	N/A	–	–
Systematic ESG Process	-0.185	21.4%	N/A	–	–
ESG Reporting	-0.367	14.3%	N/A	–	–

Comment. Cluster 3 is distinguished by a **high share of ESG / impact frameworks embedded in the fund legal documentation** (high Share Contractual) and a systematic presence of ESG analysis and risk integration, but with **little reporting** and few imposed processes. The profile is compatible with a **selective** contractualization, oriented toward "structuring" clauses rather than reporting obligations.

Transversal module — Graphical representation (MCA)



Module: Sustainable investment objective

Sustainable investment objective (52 funds)

The analysis produces a partition of the sample into 3 groups among the 52 funds concerned.

Cluster 1: Mixed focus (ENV & SOC), non-contractual, support-oriented

- Size: 14 (26.9%)
- Description: "ESG roadmap" funds. Broad sustainability focus, without formal intentionality in the fund legal documentation, non-contractual, support-oriented.

Cluster 2: Social focus without explicit target, contractualized, solutions-oriented

- Size: 12 (23.1%)
- Description: Sustainable objective and approach 100% contractualized, social (and environmental) focus, solutions-oriented. No quantified overall targets.

Cluster 3: Environmental focus, contractualized with targets, mixed strategy, combining financing & support

- Size: 26 (50%)
- Description: Environmental focus, contractualized and intentional objective, sometimes supported by frameworks, mix of financial and operational additionality.

Sustainable investment objective: Cluster 1 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 14: Sustainable investment objective — Cluster 1: signature variables (over-/under-represented)

Variable	score	p_c
Environmental objective	0.024	85.7%
Has Any Target	-0.096	0.0%
Social objective	-0.241	21.4%
Impact reporting	-0.688	0.0%
Impact analysis	-0.729	0.0%
Process IMM	-0.804	7.1%
Impact intentionality	-0.824	7.1%
Financial Additionality	–	14.3%
Operational Additionality	–	85.7%
Mixed Additionality	–	0.0%

Comment. Cluster 1 is characterized by a **mixed orientation (environmental & social)** (high Environmental objective: p_c = 85.7%, close to the average of the other clusters; significant Social objective: p_c = 21.4%, -24.1 points relative to the average of the other clusters), but with weak formal contractualization around that sustainable objective: no overall target backed by a framework, and no commitment to conduct impact reporting or specific impact analysis during the investment period (p_c = 0%). A minority of vehicles in this cluster (p_c = 7.1%) formalize their positive-impact intent and embed their IMM process in the fund legal documentation.

This cluster therefore groups funds displaying a general sustainable objective that is not contractualized at the same level as financial issues and is pursued mainly through support to portfolio companies (85.7% of the funds in this cluster have an operational additionality strategy).

Sustainable investment objective: Cluster 2 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 15: Sustainable investment objective — Cluster 2: signature variables (over-represented)

Variable	**score**	p_c
Social objective	0.688	83.3%
Process IMM	0.589	100.0%
Impact intentionality	0.568	100.0%
Impact reporting	0.562	83.3%
Impact analysis	0.396	75.0%
Has Any Target	-0.096	0.0%
Environmental objective	-0.262	66.7%
Financial Additionality	–	75.0%
Operational Additionality	–	8.3%
Mixed Additionality	–	16.7%

Comment. Cluster 2 corresponds to a **highly structured "social impact" orientation**: the IMM process and impact intentionality are systematically formalized in clauses of the fund legal documentation ($p_c = 100\%$). The majority of funds in this cluster commit to conducting specific analysis regarding their impact objective at the time of investment ($p_c = 75\%$) and to impact reporting ($p_c = 83.3\%$) in their fund legal documentation. The fact that "Has Any Target = 0" nevertheless indicates a contractualized **absence of explicit targets**.

The majority of these funds contributes to their sustainable objective through a **financial or mixed additionality strategy**, i.e. financing solutions ($p_c = 91.7\%$).

Sustainable investment objective: Cluster 3 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 16: Sustainable investment objective — Cluster 3: signature variables (over-represented)

Variable	score	p_c
Impact analysis	0.333	70.8%
Impact intentionality	0.256	79.2%
Environmental objective	0.238	100.0%
Process IMM	0.214	75.0%
Has Any Target	0.192	19.2%
Impact reporting	0.125	54.2%
Social objective	-0.447	7.7%
Financial Additionality	–	80.7%
Operational Additionality	–	15.4%
Mixed Additionality	–	3.9%

Comment. Cluster 3 corresponds to funds with an **environmental-impact** orientation ($p_c = 100\%$) formalized in their fund legal documentation: intentionality and IMM process are largely embedded ($p_c = 79.2\%$ & $p_c = 75\%$), and the majority commit to conducting specific analysis concerning their impact objective at the time of investment ($p_c = 70.8\%$).

This is the only cluster in which some funds have an impact target determined with the help of a market framework, although this remains a minority of vehicles ($p_c = 19.2\%$).

The majority of these funds contributes to their sustainable objective through a **financial or mixed additionality strategy**, i.e. financing solutions ($p_c = 84.6\%$).

Sustainable investment objective module — Graphical representation (MCA)

Sustainable investment objective — Strategy distribution

Table 17: Distribution (columns in % of column total) — Sustainable objective clusters and additionality, by strategy (BO/VC/Infra)

Cluster	N	BO	VC	Infra
Absent/NA	37	30 (57%)	7 (24%)	0 (0%)
Mixed focus, non-contractual, support-oriented	14	11 (21%)	2 (7%)	1 (14%)
Contractual social focus, no target, solutions-oriented	12	5 (9%)	7 (24%)	0 (0%)
Contractual environmental focus, support and solutions-oriented	26	7 (13%)	13 (45%)	6 (86%)
Additionality				
Absent/NA	37	30 (57%)	7 (24%)	0 (0%)
Mix	3	2 (4%)	1 (3%)	0 (0%)
Financial	32	5 (9%)	21 (72%)	6 (86%)
Operational	17	16 (30%)	0 (0%)	1 (14%)
Total	89	53 (100%)	29 (100%)	7 (100%)

Comment. Among buyout funds, 23 funds (43%) indicate that they contribute to a sustainable objective. Among them, half formalize this commitment in a clause of the fund legal documentation (12, 22%). VC and infrastructure funds more frequently have sustainable investment objectives (respectively 76% and 100%), with a predominance of environmental issues for infrastructure funds (100%). Although there is obviously a self-selection bias in our sample, we also observe that 100% of VC and infrastructure funds with a sustainable objective adopt a financial or mixed additionality strategy. This suggests that these funds contractually commit to channeling their capital toward impact assets, consistent with (i) the idea that these strategies have access to a pool of such assets and (ii) for VC funds with particularly diversified portfolios, they do not necessarily have the resources required to deploy an operational additionality strategy. Conversely, the majority of buyout funds rely on an operational or mixed additionality strategy (78%). This suggests that these funds put in place support processes in order to pursue their sustainable objective. This is consistent with (i) the idea that buyout funds have access to a smaller pool of impact assets and (ii) the fact that they operate in a favorable engagement context thanks to long holding periods and more concentrated portfolios. Finally, we recover 100% of impact funds and Article 9 funds within the clusters that contractually formalize their intentionality. It is noteworthy that there is substantial heterogeneity within Article 8 funds, half of which do not specify that they contribute to a sustainable objective.

Sustainable investment objective — Distribution by strategy, SFDR, and impact

Table 18: Distribution of sustainable objective clusters by SFDR and impact status (columns in % of column total)

Cluster	N	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
Absent/NA	37	0 (0%)	4 (100%)	13 (100%)	20 (48%)	0 (0%)
Mixed focus, non-contractual, support-oriented	14	0 (0%)	0 (0%)	0 (0%)	14 (33%)	0 (0%)
Social focus, solutions-oriented	12	9 (41%)	0 (0%)	0 (0%)	0 (0%)	12 (40%)
ENV focus, support & solutions-oriented	26	13 (59%)	0 (0%)	0 (0%)	8 (19%)	18 (60%)
Total	89	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

Comment. We again recover 100% of impact funds and Article 9 funds within the clusters that contractually formalize their intentionality. It is noteworthy that there is substantial heterogeneity among Article 8 funds, half of which do not pursue a sustainable objective and are instead positioned either on mixed issues without a contractual commitment or on environmental issues.

Module: Extra-financial committee

EF Committee (30 funds)

The analysis produces a partition of the sample into 2 groups among the 30 funds concerned.

Cluster 1: controlling EF Committee

- Size: 20 (60%)
- Description: EF Committee endowed with extensive responsibilities and effective rights in investment oversight, often with an independent component.

Cluster 2: symbolic EF Committee

- Size: 10 (40%)
- Description: EF Committee with a mainly advisory or symbolic role, with few decision-making levers.
- Signature variables: choice of beneficiary organizations for unpaid carried,

EF Committee: discriminating variables

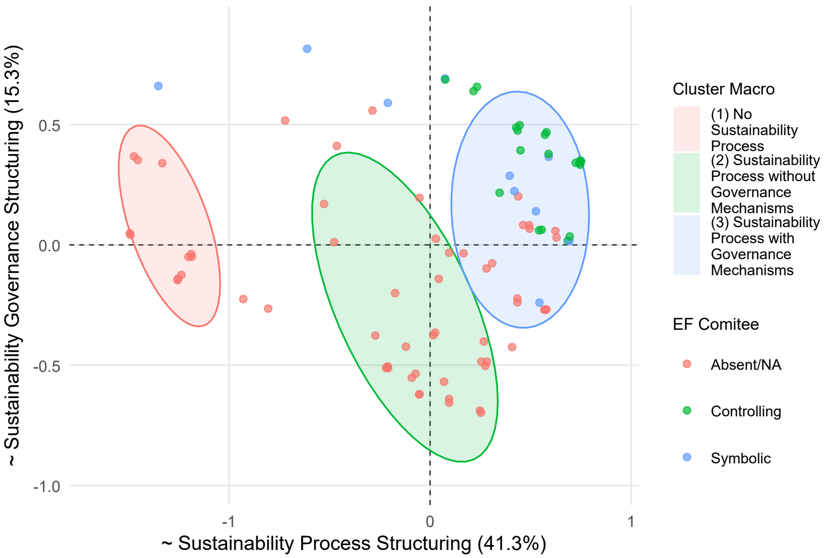
Table 19: Discriminant variables of the micro clustering — EF Committee

Variable	High cluster	Low cluster	p_{haut}	p_{bas}	Δ
Decision: EF KPIs monitored	1	2	100.0%	16.7%	83.3%
Decision: EF KPI targets	1	2	100.0%	16.7%	83.3%
Decision: Measurement validation	1	2	72.2%	0.0%	72.2%
Decision: Change in method & KPIs	1	2	61.1%	0.0%	61.1%
Decision: EF score weighting	1	2	72.2%	16.7%	55.6%
Committee: Independent members	1	2	72.2%	41.7%	30.6%
Decision: Beneficiary organizations	2	1	33.3%	11.1%	22.2%
Decision: Impact orientation	2	1	25.0%	5.6%	19.4%
Decision: Investment eligibility	2	1	33.3%	22.2%	11.1%
Decision: EF carried vesting	1	2	16.7%	8.3%	8.3%
Committee: LPs	1	2	61.1%	58.3%	2.8%
Decision: OTI measurement	1	2	11.1%	8.3%	2.8%

Comment. Funds in Cluster 1, **"Controlling Committee"**, assign **more decisions to EF Committees**, especially around the **EF KPIs monitored at the portfolio-company level**: their definition (p_c = 100% for Cluster 1, +83.3 pts vs. Cluster 2), the associated targets (p_c = 100% for Cluster 1, +83.3 pts vs. Cluster 2), their measurement (p_c = 72.2% for Cluster 1, +72.2 pts vs. Cluster 2), and their weighting in the fund's extra-financial score (p_c = 72.2% for Cluster 1, +55.6 pts vs. Cluster 2). Committees in funds from Cluster 1, "Controlling Committee", also more often grant voting seats to independent members (p_c = 72.2% for Cluster 1, +30.6 pts vs. Cluster 2).

By contrast, committees in funds from Cluster 2, **"Symbolic Committee"**, more often decide on general impact orientations (p_c = 25% for Cluster 2, +19.4 pts vs. Cluster 1), the choice of beneficiary organizations for EF carried not vested to the management company (p_c = 33.3% for Cluster 2, +22.2 pts vs. Cluster 1), or the general eligibility of the investment with respect to the impact strategy (p_c = 33.3% for Cluster 2, +11.1 pts vs. Cluster 1).

EF Committee module — Graphical representation (MCA)



EF Committee — Distribution by strategy, SFDR, and impact

Table 20: Distribution of EF Committee clusters by strategy (columns in % of column total)

Cluster	N	BO	VC	Infra
Absent/NA	59	43 (81%)	12 (41%)	4 (57%)
Controlling	20	7 (13%)	11 (38%)	2 (29%)
Symbolic	10	3 (6%)	6 (21%)	1 (14%)
Total	89	53 (100%)	29 (100%)	7 (100%)

Table 21: Distribution of EF Committee clusters by SFDR and impact status (columns in % of column total)

Cluster	N	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
Absent/NA	59	5 (23%)	4 (100%)	13 (100%)	33 (79%)	9 (30%)
Controlling	20	15 (68%)	0 (0%)	0 (0%)	2 (5%)	18 (60%)
Symbolic	10	2 (9%)	0 (0%)	0 (0%)	7 (17%)	3 (10%)
Total	89	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

Comment. Analysing the distribution of *EF Committee* clusters across strategies is not necessarily the most informative exercise given the sample size and the fact that the presence of an EF Committee may be strongly correlated with factors other than strategy, especially the fund's sustainable objective. In particular, we have seen that VC and infrastructure funds in our sample more often display this type of objective, which may also explain why they more frequently have an EF Committee. In that respect, it is more informative to examine this mechanism against the fund's additionality strategy, which is done on the next slide.

In these tables, the main point is the substantial heterogeneity among impact funds and among Article 8 and 9 SFDR funds. 77% of impact funds (17) in our sample have an EF Committee, almost systematically in the controlling form (15 funds). We observe a clear distinction between Article 8 and Article 9 funds on this issue. 79% of Article 8 funds do not have this mechanism, and 95% either do not have it or have it only in the symbolic form. By contrast, within Article 9 funds, we observe heterogeneity with majority use of the mechanism (70%), and especially in its controlling dimension (60%).

EF Committee — Distribution by additionality profiles and types of sustainable objective

Table 22: Distribution of EF Committee clusters by additionality strategy (columns in % of column total)

Cluster	N	Financial	Operational	Both
Absent/NA	59	12 (38%)	12 (71%)	0 (0%)
Controlling	20	14 (44%)	4 (24%)	2 (67%)
Symbolic	10	6 (19%)	1 (6%)	1 (33%)
Total	89	32 (100%)	17 (100%)	3 (100%)

Table 23: Distribution of EF Committee clusters by type of sustainable investment objective (columns in % of column total)

Cluster	N	Absent/NA	ENV focus, support & solutions	Mixed focus, non-contractual, support-oriented	Social focus, solutions-oriented
Absent/NA	59	35 (95%)	10 (38%)	11 (79%)	3 (25%)
Controlling	20	0 (0%)	11 (42%)	2 (14%)	7 (58%)
Symbolic	10	2 (5%)	5 (19%)	1 (7%)	2 (17%)
Total	89	37 (100%)	26 (100%)	14 (100%)	12 (100%)

Comment. We would expect the EF Committee mechanism to be particularly linked to the pursuit of a sustainable objective and to its modalities. Pursuing an additional objective may generate tensions in decision-making at key stages of the investment and support processes, on which an EF Committee may rule. These tensions may be particularly strong for operational additionality strategies (through support or engagement), and weaker for financial additionality strategies (allocation toward virtuous assets), which are characteristic of a more natural alignment between financial and societal performance.

Yet, within our sample, we observe a predominance of the presence of an EF Committee for funds deploying financial additionality (62%) or mixed additionality (100%), and a marked absence for funds with an operational additionality strategy (only 25% of funds). Across all additionality strategies, the controlling form predominates whenever this mechanism is used.

EF Committees may also play an important role when the impact objective is broad or concerns a vertical for which no measurement framework or standard targets exist. In that case, the committee may bring expertise and legitimacy to ensure the relevance of the overall IMM process. The "controlling" form is indeed predominant among funds in the "contractual social impact" cluster (58%), but also among those in the "contractual environmental impact" cluster (42%), for which one might have expected the presence of measurement frameworks and market targets to make this form less necessary. This may reflect strategies focused on innovative themes, the acquisition of specific expertise from committee members, or the use of this mechanism as a signal to stakeholders.

Module: Extra-financial carried interest

EF Carried Interest (42 funds)

The analysis produces a partition of the sample into 4 groups among the 42 funds concerned.

Cluster 1: finely calibrated EF carrieds based on specific outcome targets

- Size: 15 (35.7%)
- Description: specific outcome targets at the portfolio-company level, linear mechanism, material threshold effects (trigger at 34% of the EF score on average, full vesting at 84% on average), high indexation (44.3%).

Cluster 2: EF carrieds indexed to global outcome targets

- Size: 10 (23.8%)
- Description: global outcome targets at the portfolio level, linear mechanism without threshold effects (trigger threshold close to 0 and full-vesting threshold close to 100), medium-high indexation (36%).

Cluster 3: stepwise EF carrieds with mixed global targets

- Size: 5 (11.9%)
- Description: stepwise carried.

Cluster 4: generic EF carried

- Size: 12 (28.6%)
- Description: means-based targets at the portfolio-company level, linear without threshold effects, low indexation (16%).

EF Carried Interest: Cluster 1 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 24: EF carried interest — Cluster 1: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctl	moy. globale
Target Compensation	0.800	93.3%	N/A	–	–
Weighting by Invested Amount	0.578	93.3%	N/A	–	–
Carried Outcome Targets	0.472	93.3%	N/A	–	–
EF Carried Trigger Score	0.211	48.7%	medium	50.0%	30.0%
EF Carried Amount	0.152	44.3%	medium	61.9%	33.0%
EF Carried Paid at Trigger	0.120	34.0%	medium	66.7%	21.0%
Linear EF Carried	0.000	66.7%	N/A	–	–
EF Carried Vesting Cap	-0.081	84.0%	medium	38.1%	91.0%
Global Carried Targets	-0.494	6.7%	N/A	–	–

Comment. Cluster 1 corresponds to **demanding and "fine-tuned"** EF carried. The indexed carried amount is above the average of the other clusters and lies at the 61st percentile in the sample (44.3% on average).
These funds are characterized by indexation to **outcome targets** ($p_c = 93.3\%$, +47.2 pts relative to the average of the other clusters), defined granularly **portfolio company by portfolio company** ($p_c = 93.3\%$), with possible compensation across targets or lines ($p_c = 93.3\%$, +80 pts relative to the average of the other clusters).
The vesting parameters create material threshold effects for these funds: on average, EF carried starts vesting from an EF score of 48.7% (+21.1 pts relative to the average of the other clusters), 44.3% of EF carried is paid at that score, and it is fully vested when an EF score of 84% is reached (-8.1 pts relative to the average of the other clusters). This generally reflects rather ambitious targets, of which reaching 84% on average is considered sufficient by investors and the management company.

EF Carried Interest: Cluster 2 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 25: EF carried interest — Cluster 2: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctl	moy. globale
Global Carried Targets	0.483	80.0%	N/A	–	–
Linear EF Carried	0.444	100.0%	N/A	–	–
Carried Outcome Targets	0.428	90.0%	N/A	–	–
EF Carried Vesting Cap	0.106	98.0%	medium	42.9%	91.0%
Target Compensation	0.089	40.0%	N/A	–	–
EF Carried Amount	0.052	36.8%	medium	59.5%	33.0%
EF Carried Paid at Trigger	-0.253	6.0%	medium	64.3%	21.0%
EF Carried Trigger Score	-0.291	11.0%	medium	47.6%	30.0%
Weighting by Invested Amount	-0.667	0.0%	N/A	–	–

Comment. Cluster 2 probably corresponds to the most demanding carried mechanism in the sample. The indexed amount is slightly above the average of the other clusters and lies in the upper part of the sample distribution (36.8%).
It is generally indexed to outcome targets ($p_c = 90\%$), defined globally ($p_c = 80\%$), applied to the whole portfolio ($p_c = 80\%$), with average-level compensation across lines ($p_c = 40\%$) and weighting systematically by portfolio company ($p_c = 100\%$).
The vesting parameters are generic and do not create threshold effects. The carried is systematically linear ($p_c = 100\%$), and the trigger parameters are broad: triggering at an average EF score of 11% with payment of 6% of the EF carried, and full vesting at 98% of the EF score.

EF Carried Interest: Cluster 3 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 26: EF carried interest — Cluster 3: signature variables (over-/under-represented)

Variable	score	p_c	p_tag	pctl	moy. globale
Global Carried Targets	0.483	80.0%	N/A	–	–
EF Carried Paid at Trigger	0.467	60.0%	high	92.9%	21.0%
EF Carried Trigger Score	0.389	62.0%	high	92.9%	30.0%
EF Carried Amount	0.014	34.0%	medium	57.1%	33.0%
Weighting by Invested Amount	-0.133	40.0%	N/A	–	–
EF Carried Vesting Cap	-0.134	80.0%	medium	38.1%	91.0%
Carried Outcome Targets	-0.239	40.0%	N/A	–	–
Target Compensation	-0.444	0.0%	N/A	–	–
Linear EF Carried	-0.889	0.0%	N/A	–	–

Comment. Cluster 3 groups funds with a **stepwise vesting mechanism for EF carrieds** ($p_c = 0\%$ for the "Linear" variable). These funds tend to index EF carried to means-based targets ($p_c = 60\%$, +23.9 pts relative to the average of the other clusters), defined globally ($p_c = 80\%$, +48.3 pts relative to the average of the other clusters), with no possible compensation ($p_c = 0\%$). For these funds, the trigger threshold is high (EF score of 62% on average, 92nd percentile), with an equally high initial payment (60%, 92nd percentile). The full-vesting threshold is, however, below the average of the other clusters (80%, -13.4 pts relative to the average of the other clusters).

This reflects a coercive, threshold-based logic rather than the continuous incentive induced by the linear vesting terms of the other clusters.

EF Carried Interest: Cluster 4 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` *simply indicates the share of funds with a value lower than or equal to p_c. The* `p_tag` *label provides a synthetic reading in three levels: low, medium, or high.*

Table 27: EF carried interest — Cluster 4: signature variables (over-/under-represented)

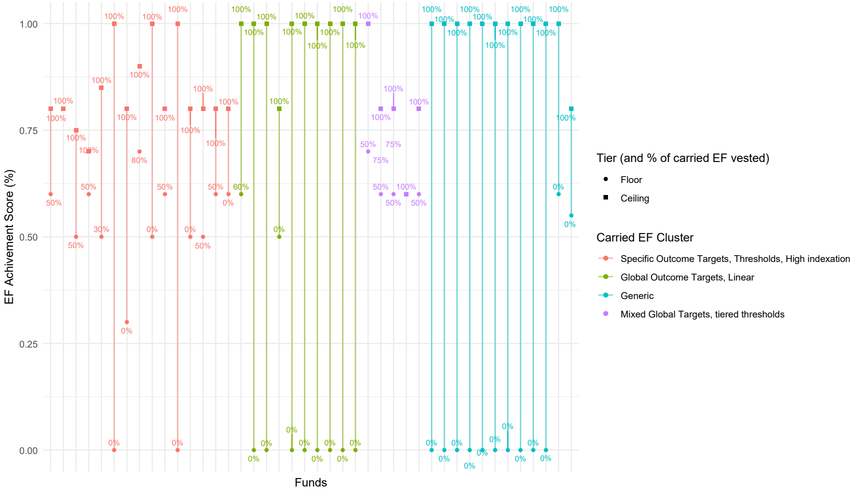
Variable	score	p_c	p_tag	pctl	moy. globale
Linear EF Carried	0.444	100.0%	N/A	–	–
Weighting by Invested Amount	0.222	66.7%	N/A	–	–
EF Carried Vesting Cap	0.110	98.3%	medium	42.9%	91.0%
EF Carried Amount	-0.217	16.7%	low	26.2%	33.0%
EF Carried Trigger Score	-0.310	9.6%	medium	47.6%	30.0%
EF Carried Paid at Trigger	-0.333	0.0%	medium	64.3%	21.0%
Target Compensation	-0.444	0.0%	N/A	–	–
Global Carried Targets	-0.472	8.3%	N/A	–	–
Carried Outcome Targets	-0.661	8.3%	N/A	–	–

Comment. Cluster 4 corresponds to the simplest EF carried configuration in terms of indexation and vesting terms. Within this cluster, the indexed share of carried is 16.7% on average (-21.7 pts relative to the average of the other clusters).

The EF carried of these funds is indexed to means-based targets ($p_c = 91.7\%$), defined specifically for each portfolio company ($p_c = 91.7\%$), with no possible compensation across lines or targets ($p_c = 0\%$).

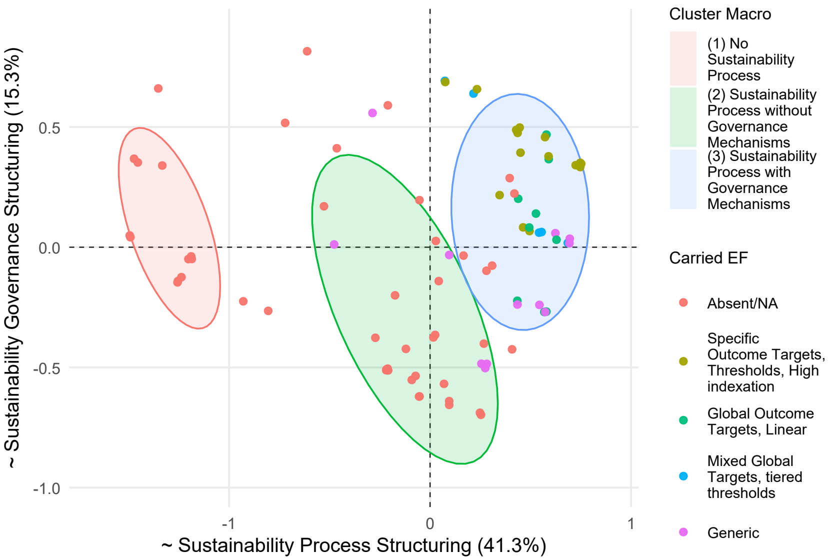
The vesting terms are generic. Vesting is linear with respect to the EF score ($p_c = 100\%$), with broad trigger and full-vesting thresholds (respectively 9.6% of the EF score and 98.3% of the EF score on average).

EF Carried Interest — Representation of the EF carried vesting mechanism by modality



Comment. This figure represents the carried vesting mechanism of funds in the sample. On the horizontal axis, each vertical line represents a fund whose carried is indexed to EF targets, with the trigger (*floor*) and full-vesting (*ceiling*) thresholds expressed as a function of the attainment of the fund's EF score. Next to each threshold is indicated the % of EF carried vested when it is reached. A continuous line indicates a linear variation between thresholds.

EF Carried Interest — Graphical representation (MCA)



EF Carried Interest — Distribution by strategy, SFDR, and impact

Table 28: Distribution of EF carried clusters by strategy (columns in % of column total)

Cluster	N	BO	VC	Infra
Absent/NA	47	32 (60%)	14 (48%)	1 (14%)
Specific outcome targets, thresholds, high indexation	15	5 (9%)	8 (28%)	2 (29%)
Global outcome targets, linear	10	5 (9%)	4 (14%)	1 (14%)
Mixed global targets, steps	5	2 (4%)	2 (7%)	1 (14%)
Generic	12	9 (17%)	1 (3%)	2 (29%)
Total	89	53 (100%)	29 (100%)	7 (100%)

Table 29: Distribution of EF carried clusters by SFDR and impact status (columns in % of column total)

Cluster	N	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
Absent/NA	47	2 (9%)	4 (100%)	13 (100%)	28 (67%)	2 (7%)
Specific outcome targets, thresholds, high indexation	15	14 (64%)	0 (0%)	0 (0%)	0 (0%)	15 (50%)
Global outcome targets, linear	10	2 (9%)	0 (0%)	0 (0%)	2 (5%)	8 (27%)
Mixed global targets, steps	5	2 (9%)	0 (0%)	0 (0%)	2 (5%)	3 (10%)
Generic	12	2 (9%)	0 (0%)	0 (0%)	10 (24%)	2 (7%)
Total	89	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

Comment. Like the EF Committee, EF carried should be linked to the fund's sustainable objective and to its modalities. In our sample, infrastructure and VC funds are those that most often adopt indexed carried (respectively 86% - 6 funds, and 52% - 15 funds). As noted above, they are also the funds that most frequently pursue a sustainable objective in our sample. Buyout funds use the mechanism relatively less (40%, 21 funds), which is consistent with the fact that they are also those least frequently pursuing a sustainable objective in our sample.

The distribution by SFDR classification and impact status yields the same result as for the *EF Committee* mechanism. The majority of impact funds use indexed carried (91%, 20 funds), most often in its most demanding modalities, with outcome targets (64% and 9%). Article 6 and unclassified funds have no indexed carried. We also observe heterogeneity among Article 8 and 9 funds. Among Article 8 funds, 44% use the mechanism, often in a generic form (24%). Among Article 9 funds, 93% have indexed carried, with dispersed modalities: the most demanding modalities based on outcome targets predominate (77%), but lighter modalities also exist.

EF Carried Interest — Distribution by additionality profiles and types of sustainable objective

Table 30: Distribution of EF carried clusters by additionality strategy (columns in % of column total)

Cluster	N	Financial	Operational	Both
Absent/NA	47	11 (34%)	4 (24%)	0 (0%)
Specific outcome targets, thresholds, high indexation	15	13 (41%)	1 (6%)	1 (33%)
Global outcome targets, linear	10	4 (12%)	5 (29%)	1 (33%)
Mixed global targets, steps	5	3 (9%)	2 (12%)	0 (0%)
Generic	12	1 (3%)	5 (29%)	1 (33%)
Total	89	32 (100%)	17 (100%)	3 (100%)

Table 31: Distribution of EF carried clusters by type of sustainable investment objective (columns in % of column total)

Cluster	N	Absent/NA	ENV focus, support & solutions	Mixed focus, non-contractual, support-oriented	Social focus, solutions-oriented
Absent/NA	47	32 (86%)	8 (31%)	6 (43%)	1 (8%)
Specific outcome targets, thresholds, high indexation	15	0 (0%)	9 (35%)	0 (0%)	6 (50%)
Global outcome targets, linear	10	0 (0%)	5 (19%)	2 (14%)	3 (25%)
Mixed global targets, steps	5	0 (0%)	1 (4%)	2 (14%)	2 (17%)
Generic	12	5 (14%)	3 (12%)	4 (29%)	0 (0%)
Total	89	37 (100%)	26 (100%)	14 (100%)	12 (100%)

Comment. As with the *EF Committee* mechanism, we expect the *EF Carried* mechanism to be particularly linked to the pursuit of a sustainable objective and to its modalities. Pursuing an additional objective may generate tensions in decision-making at key stages of the investment and support processes, and the presence of a formal incentive mechanism may therefore influence the GP's decision. These tensions may be particularly strong for operational additionality strategies (through support or engagement), and weaker for financial additionality strategies (allocation toward virtuous assets), which are characteristic of a more natural alignment between financial and societal performance.

Our observations are consistent with this hypothesis. Funds with an operational or mixed additionality strategy more frequently index their carried to EF targets than funds with a financial additionality strategy (respectively 76% and 100%, vs. 66%). For these funds, the indexation modalities are heterogeneous (35% on outcome targets, 45% on means-based or mixed targets). 66% of funds with a financial additionality strategy have indexed carried, mainly on outcome targets (53%), which may suggest that the correlation between financial and societal performance, if it exists, is not perfect.

The decomposition of *Sustainable Objective* clusters within our *EF Carried* clusters highlights substantial heterogeneity in the use of this mechanism. First, 5 funds that do not state that they pursue a sustainable objective nevertheless have indexed carried in the generic form (14%). Second, no single profile stands out for the other clusters. It is noteworthy that funds with a mixed focus based on support are those using this mechanism the least (43% do not use it), even though this is probably the modality in which tensions between financial and social performance are strongest.

Module: Impact Management & Measurement (IMM)

IMM structuring (70 funds)

The analysis produces a partition of the sample into 3 groups among the 70 funds concerned.

Cluster 1: limited IMM structuring

- Size: 31 (44.3%)
- Description: weakly structured IMM mechanisms, systematically below the average on all dimensions.

Cluster 2: fully developed IMM structuring

- Size: 21 (30%)
- Description: the most complete mechanism in the sample, with validation by an independent committee, independent measurement, and contractual integration (shareholders' agreements, management packages). Although this cluster gathers the funds with the most complete structuring in the sample, it also conceals significant disparities: only 67% record the targets in contractual documentation, 31% use a framework to define the targets, and only 24% include them in managers' management packages.

Cluster 3: intermediate IMM structuring

- Size: 18 (25.8%)
- Description: intermediate position, marked by the use of frameworks to define targets and an independent measurement process.

IMM: Cluster 1 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 32: IMM targets — Cluster 1: signature variables (over-/under-represented)

Variable	**score**	p_c
IMM: Doc Transaction	-0.028	50.0%
IMM: Exit Management	-0.082	6.5%
IMM: ManPack	-0.191	8.1%
IMM: Frameworks	-0.289	3.2%
IMM: EF Committee Validation	-0.500	0.0%
IMM: Independent Measurement	-0.929	0.0%

Comment. Cluster 1 corresponds to the funds displaying the least structured IMM process around sustainable targets. Half of the funds in this cluster integrate those targets into transaction documentation ($p_c = 50\%$), at least in aspirational form, but the other elements of the IMM process (frameworks, ManPack, etc.) are very marginal or absent.

IMM: Cluster 2 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

Table 33: IMM targets — Cluster 2: signature variables (over-represented)

Variable	**score**	p_c
IMM: EF Committee Validation	1.000	100.0%
IMM: Independent Measurement	0.357	85.7%
IMM: Doc Transaction	0.222	66.7%
IMM: Exit Management	0.178	23.8%
IMM: Frameworks	0.127	31.0%
IMM: ManPack	0.045	23.8%

Comment. Cluster 2 corresponds to the funds displaying the most structured IMM process around the extra-financial targets assigned to portfolio companies. All the funds in this cluster have these targets validated by an EF Committee ($p_c = 100\%$), and the other practices are systematically above the average of the other clusters (*score* $\in [0.05; 1]$). It is noteworthy that even among these funds, the use of frameworks to define targets, the indexation of the ManPack to these targets, and the consideration of their continuity at exit remain rare (respectively $p_c = 31\%$, $p_c = 23.8\%$, and $p_c = 23.8\%$).

IMM: Cluster 3 (signatures)

Reading note: p_c corresponds to the average value of the variable within the cluster, or to the proportion of funds taking value 1 for a binary variable. For standardized continuous variables (e.g., Functions Covered), p_c is reported as a centered z-score and can therefore be negative. The score measures the gap between this profile and the simple average of the other clusters, excluding the cluster considered. For continuous variables, `pctl` simply indicates the share of funds with a value lower than or equal to p_c. The `p_tag` label provides a synthetic reading in three levels: low, medium, or high.

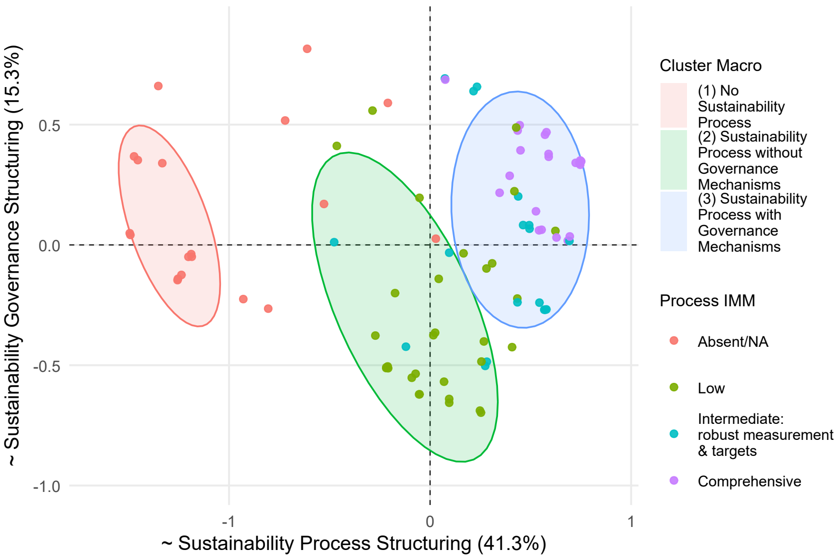
Table 34: IMM targets — Cluster 3: signature variables (over-/under-represented)

Variable	score	p_c
IMM: Independent Measurement	0.571	100.0%
IMM: Frameworks	0.162	33.3%
IMM: ManPack	0.146	30.6%
IMM: Exit Management	-0.096	5.6%
IMM: Doc Transaction	-0.194	38.9%
IMM: EF Committee Validation	-0.500	0.0%

Comment. Cluster 3 corresponds to funds displaying an intermediate IMM process. Funds in this cluster do not have their sustainable targets validated by an EF Committee ($p_c = 0\%$), even though they may have one. They are, however, more likely to define these targets using market frameworks ($p_c = 33.3\%$, +16.2 pts relative to the average of the other clusters) and to include them in managers' Management Packages ($p_c = 30.6\%$, +14.6 pts relative to the average of the other clusters). All the funds in this cluster carry out independent measurement ($p_c = 100\%$) of these targets, and a significant share include them systematically in transaction documentation ($p_c = 38.9\%$).

This may reflect two realities: (i) funds that are currently structuring their sustainability and portfolio-company support approach, or (ii) funds positioned on a sustainability axis that does not require intervention by an EF Committee or equivalent, notably because they focus on a sustainability theme for which scientific or regulatory frameworks exist, such as the energy or environmental transition.

IMM module — Graphical representation (MCA)



IMM — Distribution by strategy, SFDR, and impact

Table 35: Distribution of IMM clusters by strategy (columns in % of column total)

Cluster	N	BO	VC	Infra
Absent/NA	19	11 (21%)	8 (28%)	0 (0%)
Low	31	21 (40%)	7 (24%)	3 (43%)
Intermediate: Measurement & Robust Targets	18	14 (26%)	2 (7%)	2 (29%)
Complete	21	7 (13%)	12 (41%)	2 (29%)
Total	89	53 (100%)	29 (100%)	7 (100%)

Table 36: Distribution of IMM clusters by SFDR and impact status (columns in % of column total)

Cluster	N	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
Absent/NA	19	0 (0%)	2 (50%)	10 (77%)	7 (17%)	0 (0%)
Low	31	4 (18%)	2 (50%)	3 (23%)	22 (52%)	4 (13%)
Intermediate: Measurement & Robust Targets	18	5 (23%)	0 (0%)	0 (0%)	10 (24%)	8 (27%)
Complete	21	13 (59%)	0 (0%)	0 (0%)	3 (7%)	18 (60%)
Total	89	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

Comment. Just like the *EF Committee* and *EF Carried* mechanisms, the modalities of the *IMM* mechanism should primarily be analyzed in relation to funds' *Sustainable Objective* modalities and their additionality strategy. It is nevertheless noteworthy that 28% of VC funds have no IMM process, even though these are the funds that most frequently pursue a sustainable objective. This may reflect the fact that, for some of these funds, portfolio assets are by nature "impact-native".
The analysis relative to SFDR classifications and impact status is fairly natural. Impact funds all have an IMM process and, for the most part, the most complete structuring in the sample (59%). Unclassified funds or Article 6 funds either have no IMM process or only a low-structuring version. We observe substantial heterogeneity among Article 8 and 9 funds, with nonexistent or incomplete structuring dominating among Article 8 funds (93%), and the most complete structuring dominating among Article 9 funds (60%).

IMM — Distribution by additionality profiles and types of sustainable objective

Table 37: Distribution of IMM clusters by additionality strategy (columns in % of column total)

Cluster	N	Financial	Operational	Both
Absent/NA	19	5 (16%)	0 (0%)	0 (0%)
Low	31	7 (22%)	6 (35%)	0 (0%)
Intermediate: Measurement & Robust Targets	18	6 (19%)	7 (41%)	0 (0%)
Complete	21	14 (44%)	4 (24%)	3 (100%)
Total	89	32 (100%)	17 (100%)	3 (100%)

Table 38: Distribution of IMM clusters by type of sustainable investment objective (columns in % of column total)

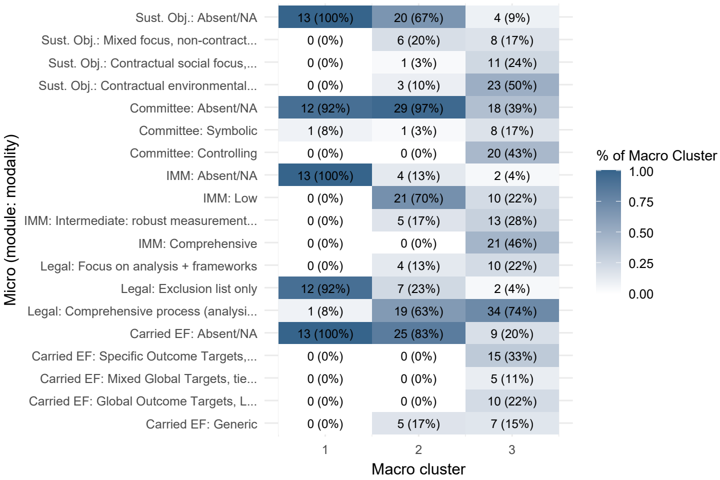
Cluster	N	Absent/NA	ENV focus, support & solutions	Mixed focus, non-contractual, support-oriented	Social focus, solutions-oriented
Absent/NA	19	14 (38%)	4 (15%)	1 (7%)	0 (0%)
Low	31	18 (49%)	5 (19%)	7 (50%)	1 (8%)
Intermediate: Measurement & Robust Targets	18	5 (14%)	5 (19%)	4 (29%)	4 (33%)
Complete	21	0 (0%)	12 (46%)	2 (14%)	7 (58%)
Total	89	37 (100%)	26 (100%)	14 (100%)	12 (100%)

Comment. These results show substantial heterogeneity in the structuring of IMM processes relative to the modalities of funds' sustainable objectives, as well as relative to their additionality strategy. A first striking result is that funds with an operational additionality strategy have lighter IMM structuring (only 24% have the "complete" modality, vs. 44% for funds with financial additionality). This may reflect (i) pursuit of objectives with highly standardized measurement frameworks and targets, (ii) a sustainability theme that is fully aligned with the financial performance of the assets ("impact-native"), or (iii) genuinely incomplete structuring relative to the strategy.

The analysis by *Sustainable Objective* modalities seems to invalidate points (i) and (ii), since funds whose *Sustainable Objective* relies only on a support strategy have weaker IMM structuring relative to the other two modalities (57% in the two least structured modalities, vs. 34% and 8% respectively).

Macro x Micro Analysis

Macro x micro analysis: visual representation



Reading note. This figure represents the decomposition of macro clusters as a function of the different modalities of the micro mechanisms. Beyond the general orientation of the funds (captured by the macro clusters), we observe substantial heterogeneity within these clusters with respect to the presence and design of the different governance mechanisms, as well as the contractualization of targets and sustainability processes. An analysis is provided on the next page.

Macro x micro analysis

Cluster 1: No sustainability approach (13 funds)
As expected, almost all the funds in this cluster do not use the different micro modules and fall within the legal cluster "Exclusions Only". One fund has a "symbolic" EF Committee or equivalent, which is consistent with the fact that these funds have no structured sustainability process. One fund also undertakes extensive legal commitments (cluster "Full Process") without stating that it has developed these capabilities operationally, likely indicating an issue driven by *LPs* and practices that still need to be developed for this vintage.

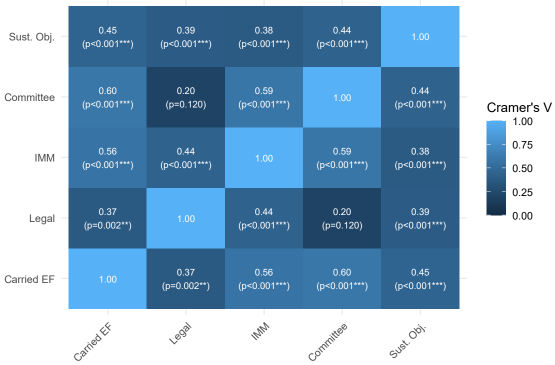
Cluster 2: Funds with a structured sustainability approach, without dedicated governance (30 funds)
Within this cluster, we begin to observe heterogeneity in process and organizational tooling. One third of the funds state that they contribute to a sustainable objective through their activity, generally a mixed one (environmental and social), without taking a formal contractual commitment (20%). By contrast, the majority fully contractualize their sustainability processes (63%), or at least the analysis component (13%), beyond a simple exclusion list (23%).
From a process perspective, the majority of funds in this cluster (87%) set sustainable targets at portfolio-company level, but have a weakly structured IMM process (70%) or one focused on measuring robust targets (17%).
Finally, the majority has neither EF carried (83%) nor an EF Committee (97%). Those that do have these mechanisms adopt the least demanding modality: linear EF carried without threshold effects, tied to means-based targets (17%), and a symbolic EF Committee (3%).

Cluster 3: Funds with a structured sustainability approach and dedicated governance (46 funds)
Funds in this cluster have a structured sustainability approach and, for the most part, use dedicated governance mechanisms (EF carried, EF Committee, explicit sustainable objective, developed IMM process).
First, 96% take contractual commitments regarding the deployment of their sustainability approach (including 74% on the full process). 91% state that they contribute to a sustainable objective, of which 74% formalize it in their fund legal documentation, more often on an environmental theme (50% vs. 24% for the social theme). In cascade, 96% assign targets to portfolio companies, but the modalities for defining and monitoring them are heterogeneous, with a fairly balanced representation of the three IMM clusters.
Second, the majority of funds implement EF carried (80%) and are accompanied by an EF Committee (61%), although this remains less common. While all EF carried modalities are represented, the two most demanding modalities (outcome targets) are the most frequently used (33% and 22%). The same is true for the EF Committee, which is more often controlling (43%).

This **Macro x Micro** analysis highlights the **overall heterogeneity** of structuring practices in sustainability matters. The **Micro x Micro** analysis will allow us to measure the joint use of the different mechanisms.

Micro x micro heatmap: statistical association

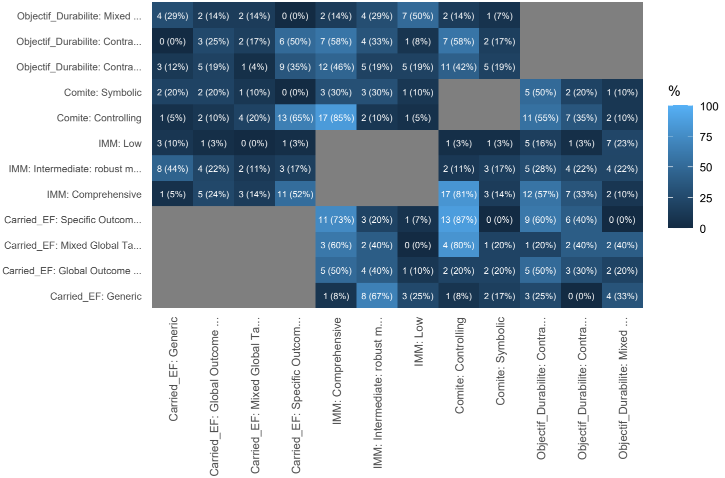
The structuring of a vehicle relies on a combination of modules (EF carried, committee, IMM, contractualization, etc.). It is therefore useful to analyze their relationship as well, in order to identify potential complementarity or substitution effects across mechanisms. This heatmap represents Cramér's V for each pair of mechanisms. This indicator measures the degree of independence between mechanism modalities. If $C_V = 1$, the mechanisms are statistically fully correlated, and the presence of one modality of one mechanism systematically implies the presence of a specific modality of the other mechanism. If $C_V = 0$, the mechanisms are fully independent. A value of 0.1, 0.3, and 0.5 is generally considered to correspond respectively to a weak, moderate, or strong association. **This value does not indicate which modalities frequently appear together, but only whether these co-occurrences are random or follow a pattern.**



Comment. Unsurprisingly, the 3 organizational mechanisms studied (Sustainable Objective, IMM Process, EF Committee, EF Carried) are associated in a moderate to strong way. This is logical for the IMM, Committee, and Carried mechanisms, since we observe that they complement one another in practice: a structured IMM process is needed to define and monitor the EF targets underpinning EF carried, and EF Committees regularly intervene around the definition of EF targets and the vesting of EF carried. The Sustainable Objective mechanism also shows a significant association with the other modules.

Micro x micro heatmap: visual representation

We analyze here the **patterns of co-occurrence** between modalities in order to identify **complementarity** and **substitution** effects underlying the statistical association observed above. This heatmap shows the **joint distribution** of the modalities of the different organizational mechanisms in the sample.



Reading. At the intersection of a row and a column, the absolute number corresponds to the number of funds displaying both modalities. The percentage corresponds to the share relative to the total number of funds having the row modality. *Example for the row generic carried and the column symbolic committee: 2 funds have these modalities, representing 17% of funds with generic carried.*

Micro x micro heatmap: analysis (1/2)

Sustainable Objective mechanism: first 3 rows. These rows restate the results previously presented mechanism by mechanism, but they enable an overall view of the sample. Several elements are striking:

- For funds with a mixed sustainable objective (non-contractual, support-oriented): frequent use of indexed carried (57% vs. 47% overall) through simple modalities based on means-based targets (43%), and a marked under-representation of modalities based on outcome targets (14% vs. 28% overall). They also less often have EF Committees than the rest of the sample, even though they rely mainly on operational additionality. While 93% set targets at portfolio-company level, their IMM structuring is heterogeneous and lighter than in the rest of the sample (79% across the two low modalities vs. 55% overall).

- 92% of funds with a contractual "social impact" orientation use an EF carried mechanism (vs. 47% for the full sample), especially in modalities based on outcome targets (75% vs. 28% for the full sample). 75% have an EF Committee (vs. 33% at sample level), generally controlling (58% vs. 22% for the sample), even though their additionality strategy is financial. Unsurprisingly, they have a more structured IMM process than average: 100% assign targets to portfolio companies (vs. 79% overall), and 58% have the most complete structuring (vs. 24% overall). It is noteworthy that, among the 3 modules, funds with a social-impact orientation display more demanding modalities than those with an environmental-impact orientation.

- For the latter, 70% have indexed carried (vs. 47% overall), in heterogeneous modalities but dominated by outcome targets (54%). 61% have an EF Committee (vs. 34% overall and 75% for social impact funds). Again, the modalities are heterogeneous, with a significant share of symbolic committees (19%, vs. 11% overall), and a lower share of controlling committees than for social impact funds (46% vs. 58%). 84% define targets with portfolio companies (vs. 79% overall, and 93% and 100% for the other sustainable objective modalities), and the IMM process modalities are also heterogeneous, although dominated by complete structuring (46%).

IMM process mechanism: next 3 rows. The IMM mechanism is mainly analyzed relative to the other mechanisms, but two points are still worth noting:

- 96% of funds with the most complete structuring also have indexed carried, mainly in the most demanding modalities based on outcome targets (76%).

- 95% of funds with the most complete structuring also have an EF Committee, most often a controlling one. This was expected since one of the steps in these complete IMM processes is validation by an independent committee.

Micro x micro heatmap: analysis (2/2)

EF Committee mechanism: just like the IMM process mechanism, committee modalities are mainly useful to consider in relation to the other mechanisms. Three points are nevertheless worth noting:

- 100% of funds with a "Controlling" committee also have carried interest indexed to extra-financial targets, mainly (75%) in the two most demanding modalities based on outcome targets. This reinforces the idea that this committee modality is complementary to the other mechanisms, and in particular helps legitimize the targets underlying EF carrieds.
- 100% of funds with a "Controlling" committee set targets for portfolio companies, and 85% also have fully developed IMM structuring. This is linked to the nature of the "Complete IMM" cluster, which includes target validation by the committee as a signature variable. Again, this suggests a complementarity effect between mechanisms.
- Only 50% of funds with a symbolic committee use EF carried. There may therefore be a substitution effect between the committee and carried for these funds.

EF Carried mechanism:

- Whatever the EF carried modality, 100% of funds with indexed carried set targets at portfolio-company level and have a process to support them. This is an expected result since EF carried is always tied to targets linked to portfolio companies. IMM structuring is nevertheless fairly heterogeneous depending on the EF carried modality: the most demanding modalities are associated with a complete IMM process (73%, 50%, 60%) or at least with an independent measurement process (20%, 40%, 40%). The "generic" modality typically comes with only a measurement process (92%), often independent (67%).
- The relationship with the use of an EF Committee is fairly heterogeneous and mainly linked to the nature of the targets. Among funds with the most demanding carried modality (specific outcome targets), 87% have a controlling committee, suggesting fairly strong complementarity between these modalities: impact committees validate and monitor the targets on which carried is indexed. This is less the case for funds with global outcome targets (20%), supporting the same idea and suggesting that when the impact theme is positioned on a theme that allows common targets across portfolio companies, with clear ambition backed by a framework, the role of an EF Committee is more limited. 75% of funds with generic carried have no committee, often because they are tied to generic targets assigned to portfolio companies.

If this analysis allows us to understand the extent to which the mechanisms are used jointly and the reasons associated with that, it remains a mechanism-by-mechanism analysis. The final step is therefore to analyze the profiles (combinations of all mechanisms) that are most dominant in our sample.

Typology: Organizational Signatures of Vehicles

Organizational signatures: methodological approach

Objective. Identify *organizational configurations* of funds (mechanisms mobilized, governance and incentive modalities) and their prevalence in the sample.

Choice of variables (non-redundancy principle).

- The signatures are based on the **modalities of the mechanisms** because they describe their effective *design*: EF carried, committee, IMM, etc.).
- Macro variables strongly correlated with micro modalities (e.g. IMM screening and measurement variables, flags for use of a mechanism) are **excluded** in order to avoid double counting and excessive weighting of the same mechanism. This also allows a more synthetic reading of the organizational signatures.
- In addition, **only the sustainability team** is added as a transversal variable: it captures an *organizational capacity* not reducible to mechanism modalities.

Step 1: Combinatorial analysis (signatures).

- Construction of signatures from the **modalities of the mechanisms**, with the *Absent/NA* modality when the mechanism is absent.
- Three reading levels: (i) *global* (distribution over the whole sample), (ii) *conditioned* by sustainable-objective modality, then (iii) *conditioned* by type of additionality strategy.

Step 2: Robust typology (clustering).

- **Why perform a last clustering analysis?** Combinatorial analysis offers an exhaustive description, but with a small sample (89 vehicles) it generates a large number of low-frequency signatures, inducing (i) fragmentation of information (many single or quasi-single cases) and (ii) difficulties in identifying empirical regularities.
- **Principle.** Estimation of a typology by clustering on the **same building blocks as the combinatorial analysis** (micro modalities + sustainability team), in order to group nearby signatures and produce an organizational typology of funds.
- **Deliverable.** A stable typology of 8 **structuring profiles**: for each cluster, (i) counts and prevalence, (ii) a *profile* of over-/under-represented modalities, and (iii) a synthetic reading of structuring logics.

Overall reading: combinatorial analysis maps configurations, while clustering enables the synthesis of the dominant profiles within a typology.

Organizational signatures: global focus

S	Sustainable_Objective	Sust. Team	IMM	Carried_EF	Committee	N	%	cumulative
1	Absent/NA	No (0)	Absent/NA	Absent/NA	Absent/NA	11	12,4%	12,4%
2	Absent/NA	Yes (1)	Low	Absent/NA	Absent/NA	9	10,1%	22,5%
3	Absent/NA	No (0)	Low	Absent/NA	Absent/NA	8	9,0%	31,5%
4	ENV focus, support & solutions	Yes (1)	Complete	Specific Outcome Targets	Controlling	5	5,6%	37,1%
5	Absent/NA	No (0)	Intermediate: Measurement & Robust Targets	Generic	Absent/NA	3	3,4%	40,4%
6	Mixed focus, support-oriented	Yes (1)	Low	Absent/NA	Absent/NA	3	3,4%	43,8%
7	Mixed focus, support-oriented	No (0)	Low	Absent/NA	Absent/NA	2	2,2%	46,1%
8	ENV focus, support & solutions	No (0)	Complete	Specific Outcome Targets	Controlling	2	2,2%	48,3%
9	Social focus, solutions-oriented	No (0)	Complete	Specific Outcome Targets	Controlling	2	2,2%	50,6%
10	Absent/NA	Yes (1)	Absent/NA	Absent/NA	Absent/NA	2	2,2%	52,8%
11	Mixed focus, support-oriented	Yes (1)	Complete	Mixed Global Targets, Steps	Controlling	2	2,2%	55,1%
12	ENV focus, support & solutions	Yes (1)	Complete	Global Outcome Targets	Controlling	2	2,2%	57,3%
13	Social focus, solutions-oriented	Yes (1)	Complete	Global Outcome Targets	Symbolic	2	2,2%	59,6%
14	Social focus, solutions-oriented	Yes (1)	Complete	Specific Outcome Targets	Controlling	2	2,2%	61,8%

Comment. There are 48 organizational signatures in total in the sample. We present the 14 associated with at least two funds, representing 61.8% of the sample. The most represented configuration corresponds to 12.4% of the sample, and only 3 configurations account for more than 5% of the sample, reflecting a fairly high heterogeneity in the organizational structures of the vehicles. As expected, funds without a sustainable objective dominate (31.5% cumulatively), since they are naturally grouped into a small number of weakly structured combinations. The following pages present an analysis by sustainable-objective modality, but here are the main takeaways for signatures without a sustainable objective:

- The most frequent signature (12.4%) represents funds with no sustainability mechanism at all.
- 17 funds (19.1%, signatures 2 and 3) have no sustainable objective but nevertheless set targets with their portfolio companies, with a weakly structured IMM process. Among them, 9 have a dedicated sustainability team.

Organizational signatures: focus by sustainable objective

Rank	Sustainable_Objective	Sust. Team	Carried_EF	IMM	Committee	N	%	Cumulative
1	Absent/NA	No (0)	Absent/NA	Absent/NA	Absent/NA	11	29,7%	29,7%
2		Yes (1)	Absent/NA	Low	Absent/NA	9	24,3%	54,1%
3		No (0)	Absent/NA	Low	Absent/NA	8	21,6%	75,7%
4		No (0)	Generic	Intermediate: Measurement & Robust Targets	Absent/NA	3	8,1%	83,8%
5		Yes (1)	Absent/NA	Absent/NA	Absent/NA	2	5,4%	89,2%
1	ENV focus, support & solutions	Yes (1)	Specific Outcome Targets	Complete	Controlling	5	19,2%	19,2%
2		No (0)	Specific Outcome Targets	Complete	Controlling	2	7,7%	26,9%
3		Yes (1)	Global Outcome Targets	Complete	Controlling	2	7,7%	34,6%
1	Mixed focus, support-oriented	Yes (1)	Absent/NA	Low	Absent/NA	3	21,4%	21,4%
2		No (0)	Absent/NA	Low	Absent/NA	2	14,3%	35,7%
3		Yes (1)	Mixed Global Targets, Steps	Complete	Controlling	2	14,3%	50,0%
1	Social focus, solutions-oriented	No (0)	Specific Outcome Targets	Complete	Controlling	2	16,7%	16,7%
2		Yes (1)	Specific Outcome Targets	Complete	Controlling	2	16,7%	33,3%
3		Yes (1)	Global Outcome Targets	Complete	Symbolic	2	16,7%	50,0%

Comment. We focus here on the signatures of funds targeting a sustainable objective. Regarding funds pursuing an environmental objective, the most represented signatures (34.6%) indicate complete structuring with advanced mechanisms: controlling committee, most complete IMM process, and carried indexed to outcome targets. One group of funds does not have a dedicated sustainability team, which may indicate that they seek this resource externally or among members of the investment team directly. This group nevertheless displays very dispersed signatures, and 34.6% (9 funds) have a weakly structured signature (two of the three mechanisms absent or in a weak modality).

The finding is similar for funds pursuing a purely social objective. The most represented signatures (50% of these funds) display high structuring modalities, sometimes without a dedicated sustainability team. Among these funds, particularly weakly structured signatures (two of the three mechanisms absent or in a weak modality) are particularly rare, since they concern only one vehicle (8%).

For funds pursuing a mixed objective, often less contractualized, the finding is more nuanced. The signatures are concentrated (3 signatures represent 50% of these funds) and reveal weak structuring with two of the three mechanisms absent or in simple modalities (35.3% of funds). One important signature reveals complete structuring (14.3%).

Beyond the modality of the sustainable objective pursued, the structuring of a vehicle may also depend strongly on the additionality strategy employed, which is the subject of the next slide.

Organizational signatures: focus by additionality strategy

S	Additionality	Carried_EF	IMM	Committee	Sust. Team	N	%	cumulative
1	Operational	Absent/NA	Low	Absent/NA	Yes (1)	3	15,0%	15,0%
2		Global Outcome Targets	Intermediate: Measurement & Robust Targets	Absent/NA	Yes (1)	2	10,0%	25,0%
3		Mixed Global Targets, Steps	Complete	Controlling	Yes (1)	2	10,0%	35,0%
4		Generic	Intermediate: Measurement & Robust Targets	Absent/NA	Yes (1)	2	10,0%	45,0%
5		Specific Outcome Targets	Complete	Controlling	Yes (1)	2	10,0%	55,0%
1	Financial	Specific Outcome Targets	Complete	Controlling	Yes (1)	5	15,6%	15,6%
2		Specific Outcome Targets	Complete	Controlling	No (0)	4	12,5%	28,1%
3		Absent/NA	Absent/NA	Absent/NA	No (0)	2	6,2%	34,4%
4		Absent/NA	Low	Absent/NA	No (0)	2	6,2%	40,6%
5		Absent/NA	Low	Absent/NA	Yes (1)	2	6,2%	46,9%

Comment. We focus here on the signatures of funds according to their additionality strategy. We observe substantial heterogeneity in the signatures of funds deploying an operational additionality strategy. 75% have a sustainability team, but the use of the other mechanisms is uneven, as shown by the 5 most dominant signatures, in which all modalities of each mechanism are represented. The absence of an EF Committee in these signatures is particularly noteworthy, given that such a committee can be a useful tool for managing trade-offs and setting ambitious targets for portfolio companies in the context of an operational additionality strategy.

Funds deploying a financial additionality strategy have more concentrated heterogeneous signatures. 9 funds (28.1%) have advanced structuring across all mechanisms, and 4 of them have no sustainability team. Again, this does not necessarily reflect a lack of structuring, but may indicate frequent use of advisory support or expertise within the investment team. Beyond these signatures, the other dominant signatures reflect weak structuring (mechanisms mostly absent or in a weakly structured modality). This is not surprising: financial additionality strategies are often characteristic of a stronger correlation between sustainable objectives and financial performance, since the underlying assets are by nature virtuous. As a result, they may require fewer mechanisms to manage potential tensions and trade-offs between the dimensions of performance, which are mechanically more closely linked. The advanced structuring of some funds nevertheless suggests that it may still be necessary to internalize specific capabilities depending on the sustainability vertical addressed, or that this may also be put in place as a signaling effect toward LPs.

Final macro typology of organizational signatures (K=8)

The analysis produces a partition of the sample into 8 organizational fund signatures. The size of our sample limits the scope of the combinatorial analysis, which highlights a number of unique signatures. We therefore conduct a final clustering analysis, this time on the precise use of the mechanisms, in order to identify the overall empirical profiles emerging from our sample. The variables used are the 4 micro mechanisms (Sustainable Objective, IMM, EF Carried, Committee), as well as the presence of a sustainability team.

Type	Nom	n	%
1	No structured approach	14	15.7
2	Environmental objective - Symbolic structuring	7	7.9
3	Social objective - Complete architecture	11	12.4
4	Environmental objective - Complete architecture	11	12.4
5	Minimal IMM process - No objective	11	12.4
6	Internal resource without governance mechanism - No objective	18	20.2
7	Mixed objective - Generic carried & independent measurement	9	10.1
8	Environmental objective - Robust incentive	8	9.0

Description of signature types (1/3)

For each variable, the tables indicate the dominant modality in the cluster. The percentage corresponds to its frequency in the cluster. The standardized residual (r) indicates how much more (or less) this modality is represented in this cluster than in the sample as a whole. In practice, a residual above 2 indicates a particularly distinctive trait of funds in the cluster.

Cluster 1 — No structured approach (n=14 ; 15.7%)

Variable	Dominant modality
IMM	Absent/NA (100%, $r = +6.37$)
Sust. Team	No (100%, $r = +2.97$)
Sust. Objective	Absent/NA (85.7%, $r = +2.56$)
EF Carried	Absent/NA (100%, $r = +2.43$)
EF Committee	Absent/NA (92.9%, $r = +1.22$)

Reading of Cluster 1

This cluster corresponds to the least structured funds in the sample. Sustainability mechanisms are almost systematically absent: no dedicated team, no EF carried, no committee, and no formalized IMM approach. These are mostly funds with no sustainable objective (85.7%). As a result, 14.3% of these funds pursue a sustainable objective without any dedicated resource or governance mechanism.

Cluster 2 — Environmental Objective - Symbolic Structuring (n=7 ; 7.9%)

Variable	Dominant modality
EF Committee	Symbolic (85.7%, $r = +5.88$)
Sust. Objective	Contractual environmental objective (85.7%, $r = +2.77$)
IMM	Absent/NA (42.9%, $r = +1.23$)
EF Carried	Absent/NA (71.4%, $r = +0.68$)
Sust. Team	Yes (71.4%, $r = +0.63$)

Reading of Cluster 2

This cluster groups funds with a contractualized environmental objective, accompanied by primarily symbolic structuring: these funds have an EF Committee that is above all symbolic, while the other mechanisms remain only weakly activated. This structuring could be sufficient for funds deploying a purely financial additionality strategy, with environmental objectives perfectly aligned with the financial performance of underlying assets.

Cluster 3 — Social Objective - Complete Architecture (n=11 ; 12.4%)

Variable	Dominant modality
Sust. Objective	Contractual social objective (81.8%, $r = +6.17$)
EF Committee	Controlling (81.8%, $r = +4.15$)
IMM	Complete (81.8%, $r = +3.98$)
EF Carried	Specific outcome targets (45.5%, $r = +2.31$)
Sust. Team	Yes (72.7%, $r = +0.85$)

Reading of Cluster 3

Funds with a contractualized social impact objective, accompanied by complete organizational structuring. They combine a contractual social objective, the most complete IMM process in the sample, a controlling EF Committee, and EF carried indexed to outcome targets assigned to portfolio companies. This seems characteristic of funds pursuing a demanding social objective, for which no standardized framework exists and whose legitimacy and monitoring require complete structuring including an EF Committee.

Description of signature types (2/3)

For each variable, the tables indicate the dominant modality in the cluster. The percentage corresponds to its frequency in the cluster. The standardized residual (r) indicates how much more (or less) this modality is represented in this cluster than in the sample as a whole. In practice, a residual above 2 indicates a particularly distinctive trait relative to the funds in the cluster.

Cluster 4 — Environmental Objective - Complete Architecture (n=11 ; 12.4%)

Variable	Dominant modality
EF Committee	Controlling (100%, $r = +5.42$)
IMM	Complete (90.9%, $r = +4.60$)
EF Carried	Specific outcome targets (72.7%, $r = +4.51$)
Sust. Objective	Contractual environmental objective (100%, $r = +4.34$)
Sust. Team	Yes (72.7%, $r = +0.85$)

Reading of Cluster 4

Equivalent of Cluster 3, for environmental impact funds. This structuring seems typically suited to vehicles deploying an operational or mixed additionality strategy, since these mechanisms enable the management of potential tensions between environmental performance and financial performance.

Cluster 5 — Minimal IMM Process (n=11 ; 12.4%)

Variable	Dominant modality
IMM	Low (100%, $r = +3.66$)
Sust. Team	No (100%, $r = +2.64$)
EF Carried	Absent/NA (100%, $r = +2.15$)
Sust. Objective	Absent/NA (72.7%, $r = +1.60$)
EF Committee	Absent/NA (100%, $r = +1.37$)

Reading of Cluster 5

Funds characterized by minimal structuring around sustainability issues, relying only on a process for measuring ESG indicators. While the majority of these funds do not pursue a sustainable objective, 3 (72.7%) do pursue a mixed or environmental objective. This raises questions about their operational capacity and the incentives they have to do so, especially if they rely on an operational additionality strategy, given that they are not structured around that objective.

Cluster 6 — Internal capacity without governance mechanism (n=18 ; 20.2%)

Variable	Dominant modality
IMM	Low (83.3%, $r = +3.49$)
Sust. Team	Yes (100%, $r = +2.66$)
EF Carried	Absent/NA (94.4%, $r = +2.43$)
Sust. Objective	Absent/NA (72.2%, $r = +2.02$)
EF Committee	Absent/NA (100%, $r = +1.76$)

Reading of Cluster 6

Cluster equivalent to Cluster 5, simply with an additional sustainability team. Again, most mechanisms are absent, which is consistent with the absence of a sustainable objective. Yet 5 funds (27.8%) do pursue such an objective. While these funds are characterized by a higher level of internal resources to develop extra-financial performance, this raises the question of the level of incentive to pursue that objective, especially if it conflicts in the short term with financial performance.

Description of signature types (3/3)

Cluster 7 — Mixed focus - generic EF carried + IMM (n=9 ; 10.1%)

Variable	Dominant modality
EF Carried	Generic (77.8%, $r = +5.25$)
IMM	Intermediate: Targets & Measurement (77.8%, $r = +3.84$)
Sust. Objective	Non-contractual mixed objective (55.6%, $r = +3.01$)
Sust. Team	No (77.8%, $r = +1.40$)
EF Committee	Absent/NA (88.9%, $r = +0.83$)

Reading of Cluster 7

Cluster of funds pursuing a mixed sustainable objective, not contractualized. This objective is supported by intermediate/weak structuring, with generic EF carried (means-based targets, defined for each portfolio company) and independent measurement of the targets linked to carried. This is a structuring that leaves the EF targets associated with carried entirely in the hands of the GP, without validation by an EF Committee or adherence to frameworks. This seems typical of funds that articulate the EF objective in an **instrumental** manner, **in the service of financial performance**.

Cluster 8 — Environmental objective - Robust incentive (n=8 ; 9.0%)

Variable	Dominant modality
EF Carried	Global outcome targets (62.5%, $r = +4.33$)
IMM	Robust intermediate (75.0%, $r = +3.44$)
Sust. Objective	Contractual environmental objective (75.0%, $r = +2.40$)
Sust. Team	Yes (87.5%, $r = +1.29$)
EF Committee	Absent/NA (100%, $r = +1.17$)

Reading of Cluster 8

This last cluster groups funds that contractualize the pursuit of a specific environmental objective at fund level. As a result, they rely mainly on a robust incentive mechanism to frame the pursuit of that objective: carried indexed to outcome targets at portfolio level, combined with an IMM process centered on the independent measurement of indicators backed by frameworks. In this configuration, the role of an EF Committee could be more limited, as potential trade-offs are foreseeable and primarily managed via EF carried — an interpretive hypothesis that remains to be verified.

Overall reading

The typology highlights a structuring continuum ranging from an almost total absence of mechanisms (Clusters 1 and 5) to complete architectures (Clusters 3 and 4), passing through several intermediate forms: symbolic structuring (Cluster 2, especially around the environmental objective), development of internal resources without dedicated governance (Cluster 6), or structuring through incentives (Clusters 7 and 8, with different levels of rigor). The observed configurations therefore differ not only in terms of degree of equipment (use of mechanisms), but also strongly in the association of specific mechanism modalities. Several stylized facts emerge from this analysis. First, some funds pursuing a sustainable objective are only weakly structured around that objective and, in particular, have weakly coercive governance (especially generic EF carried), leaving priority to financial performance: Clusters 5, 6, and 7. This may be characteristic of funds pursuing sustainable objectives only in an instrumental logic, in order to strengthen the financial performance of companies. Second, our typology shows that some GPs use fund architecture for signaling and legitimacy purposes: some funds relying on a financial additionality strategy display complete structuring, whereas lighter or more customized structuring would have been sufficient where trade-offs are limited. The use of governance mechanisms might have seemed linked to the fund's additionality strategy. A financial additionality strategy should induce fewer tensions and trade-offs between financial and sustainability objectives during the investment and support process than an extra-financial additionality strategy based on supporting assets without native impact, thus requiring fewer dedicated governance mechanisms. We discuss this point in detail in the conclusion.

Organizational typology — distribution by SFDR and impact

Table 39: Distribution of macro clusters by SFDR classification and impact status (columns in % of column total)

Cluster	N	Non Impact	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
No structured approach	14	14 (21%)	0 (0%)	2 (50%)	8 (62%)	4 (10%)	0 (0%)
Minimal IMM process – no objective	11	11 (16%)	0 (0%)	1 (25%)	3 (23%)	7 (17%)	0 (0%)
Internal resource	18	17 (25%)	1 (5%)	1 (25%)	2 (15%)	13 (31%)	2 (7%)
Mixed objective – EF carried + moderate IMM	9	8 (12%)	1 (5%)	0 (0%)	0 (0%)	8 (19%)	1 (3%)
Env. objective – symbolic structuring	7	6 (9%)	1 (5%)	0 (0%)	0 (0%)	6 (14%)	1 (3%)
Env. objective – robust incentive	8	5 (7%)	3 (14%)	0 (0%)	0 (0%)	2 (5%)	6 (20%)
Env. objective – complete architecture	11	2 (3%)	9 (41%)	0 (0%)	0 (0%)	0 (0%)	11 (37%)
Social objective – complete architecture	11	4 (6%)	7 (32%)	0 (0%)	0 (0%)	2 (5%)	9 (30%)
Total	89	67 (100%)	22 (100%)	4 (100%)	13 (100%)	42 (100%)	30 (100%)

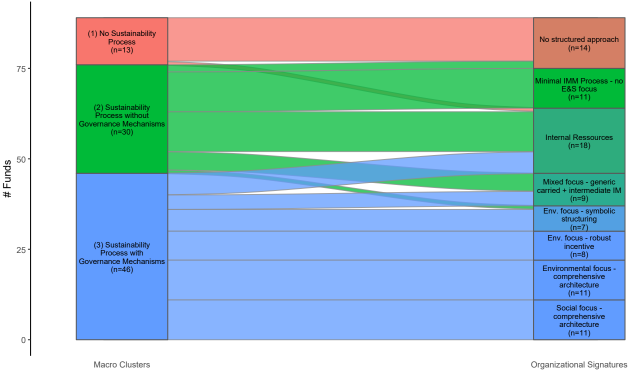
Comment. Logically, Article 6 funds and non-impact funds are concentrated mostly in the first two clusters (no structured approach or minimal IMM process), whereas Article 9 funds and impact funds are found mainly (respectively 90% and 92%) in more complete architectures, consistent with their sustainability ambitions (environmental or social objectives supported by incentive and/or governance mechanisms). Article 8 funds are more spread across the different organizational signatures, and are the only category represented across almost the entire typology. The distinction between impact and non-impact funds, and between SFDR 8 and 9 classifications, is nevertheless not fully reflected in the organizational signatures of the funds:

- 11 non-impact funds (16.4%) have an organizational signature structured around a sustainable objective,
- 4 Article 9 funds (13%) have minimal or symbolic structuring around their sustainable objective,
- and 4 Article 8 funds (10%) have complete structuring around their sustainable objective.

This reflects (i) the heterogeneity of the organizational structures supporting funds' sustainability ambitions and (ii) the fact that regulation and market frameworks relating to impact only very partially account for this organizational heterogeneity.

Organizational typology: distinction from the initial macro distribution

The following figure enables an analysis of the distribution of the initial macro-level decomposition within the final organizational typology. As a reminder, we had formed 3 clusters based on "facial" variables presented in the commercial documentation of the funds: no sustainability approach, structured sustainability process without associated governance, and structured sustainability process with associated governance.



Reading: The color of the organizational-typology boxes corresponds to a weighted blend of the colors of the macro clusters composing them, as a function of the relative share of each macro cluster in the signature considered.

Comment. Logically, almost all the funds in Macro Cluster 1 (no sustainability process) display a "No structured approach" organizational signature, while those in Macro Cluster 3 (Sustainability with governance) tend to display one of the more complete signatures.

It is particularly noteworthy that (i) there is a heterogeneous organizational reality within Macro Clusters 2 and 3, which are distributed across 5 and 6 organizational signatures respectively, and (ii) the types "Internal Resource", "Mixed Objective - EF Carried + Moderate IMM", and "Environmental Objective - Symbolic Structuring" are shared between these two clusters. **Again, this reinforces the idea of substantial organizational heterogeneity behind the display of sustainability-process structuring and the use of certain governance mechanisms.**

Conclusion

Conclusion: strong organizational heterogeneity around sustainability issues

This study does not focus only on the intentions of funds in terms of sustainability, but on their **concrete structuring** through the combination of organizational mechanisms:

- Value-creation objectives and investment policy (financial, social, environmental),
- Resources mobilized to achieve these objectives (dedicated team, development of a sustainability process),
- Incentive mechanisms (EF carried) and control mechanisms (EF Committee or equivalent) framing their implementation.

It relies on an original database built from the analysis of commercial and legal documents from 89 funds, classified in three steps using a clustering method:

- a first **macro** level, describing the general architecture of the vehicles, and leading to three broad clusters based on commercial variables (no sustainability approach, structured sustainability approach without dedicated governance, sustainability approach with dedicated governance),
- a second **micro** level, describing the effective design of the mechanisms when they are used, and bringing out several modalities (2 to 4) per mechanism,
- a final **macro** level, establishing an empirical typology of 8 organizational signatures, grouping funds according to the combination of organizational mechanisms and their modalities.

We do not seek to claim that one architecture is *in itself* superior to another. Our objective is to **map** the different architectures observed in our sample and discuss their relevance in light of the financial and sustainability objectives pursued.

→ **The central message of this study is that the development of sustainable finance in French private equity has indeed been accompanied by substantial heterogeneity in the intentionality of funds, but above all in the organizational architecture supporting those intentions.** This is a central characteristic of vehicles, but one that remains structurally little visible today and is only weakly captured by market frameworks.

First result: a structuring continuum around sustainability

The first analysis brings out **three types of funds**:

- No sustainability approach (14.6%),
- Structured approach **without dedicated governance** (33.7%),
- Structured approach **with dedicated governance** (51.7%).

A substantial share of the market (85.4% of the sample) has therefore integrated a **basic ESG process layer** (screening, support, reporting) as well as a **contractual base** (exclusion list, generic ESG integration clauses, etc.). But this surface-level structuring does not imply the explicit pursuit of a **sustainable objective**, the development of **dedicated resources** to achieve that objective, the presence of robust **governance mechanisms**, or the overall coherence of this architecture.

Table 40: Distribution of the three macro clusters by impact status and SFDR classification. Percentages are computed by column.

Cluster macro	N	Non Impact	Impact	SFDR NA	SFDR 6	SFDR 8	SFDR 9
No sustainability approach	13	13 (100%)	0 (0%)	2 (15%)	9 (69%)	2 (15%)	0 (0%)
Structured sustainability without dedicated governance	30	29 (97%)	1 (3%)	2 (7%)	4 (13%)	23 (77%)	1 (3%)
Structured sustainability with dedicated governance	46	25 (54%)	21 (46%)	0 (0%)	0 (0%)	17 (37%)	29 (63%)
Total	89	67 (75%)	22 (25%)	4 (4%)	13 (15%)	42 (47%)	30 (34%)

→ **The market is broadly committed to an ESG process and legal base, but the organizational depth of that commitment remains highly variable, already at the level of the use of sustainability-related organizational mechanisms.**

Second result: the effective design of mechanisms is heterogeneous

The modalities of organizational mechanisms are multiple, and the difference lies in a small number of variables.

- **Sustainable objective (3 modalities):** funds differ first by the **type of objective pursued** (environmental, social, mixed), their **additionality strategy** (financial, operational, mixed), and the **degree of contractual enforcement** (presence of explicit targets, intentionality formalized in the rules, impact reporting, impact analysis, and a formalized IMM process).
- **EF Committee (2 modalities):** beyond the presence of the committee, one distinguishes **controlling** committees and **symbolic** committees. The most structuring variables are the **effective rights** over the EF KPIs monitored, the associated targets, validation of measurement, change of method or indicators, weighting of the extra-financial score, and the presence of **independent members** on the committee.
- **EF Carried (4 modalities):** configurations differ according to the **nature of indexed targets** (means vs outcomes, global vs portfolio company by portfolio company), the possibility of **compensation** between targets, the **share of carried indexed**, and the vesting mechanics (linear vs tiered, vesting thresholds).
- **IMM (3 modalities):** one distinguishes **low**, **intermediate**, and **complete** structuring. The most structuring variables are the integration of targets into **transaction documentation**, the use of **frameworks**, integration into **management packages**, **committee validation**, **independent measurement**, and, more broadly, the embedding of the mechanism in steering and exit.

In other words, two funds may display the use of a similar mechanism while giving it a very different scope within the vehicle architecture depending on the modality used.

→ **What matters is not only the presence of an organizational mechanism, but its precise mechanics and what that implies in terms of incentive, control, and coercion.**

Third result: summary organizational typology

The final typology in **8 organizational signatures** constitutes the synthesis of the work carried out across all modules.

Table 41: Final typology of the 8 organizational signatures observed in the sample.

Type	Nom	n	%
1	No structured approach	14	15.7
2	Environmental objective - Symbolic structuring	7	7.9
3	Social objective - Complete architecture	11	12.4
4	Environmental objective - Complete architecture	11	12.4
5	Minimal IMM process - No objective	11	12.4
6	Internal resource without governance mechanism - No objective	18	20.2
7	Mixed objective - Generic carried & independent measurement	9	10.1
8	Environmental objective - Robust incentive	8	9.0

It shows that:

- organizational heterogeneity is **very strong**, including among the funds that appear a priori the most committed,
- the most frequent architectures are not necessarily the most complete,
- several intermediate forms rely first on **internal resources** and processes, without a truly dedicated governance mechanism.

The final typology therefore allows a finer reading than ex ante categories, because it restores the **effective combinations** of objectives, resources, and mechanisms.

→ **Beyond heterogeneity at the level of intentions, this typology reveals distinct organizational architectures.**

Fourth result: A possible symbolic role for structuring?

One can distinguish two broad additionality logics: **Financial** (allocation of capital toward assets or solutions already aligned with the sustainable objective), and **Operational** (transformation of portfolio companies through support and shareholder engagement).

In theory, one might expect more demanding structuring when the strategy relies more heavily on an operational additionality logic. The tensions between financial and extra-financial performance are potentially stronger there, especially in the short term. The organizational architecture of the fund should therefore be fine-tuned in order to frame these trade-offs, for example with an EF Committee and EF carried.

Table 42: Distribution of organizational signatures by type of additionality (column percentages)

Signature	NA	Fin.	Op.	Mix
No structured approach	12 (32%)	2 (6%)	0 (0%)	0 (0%)
Minimal IMM process – no objective	8 (22%)	2 (6%)	1 (6%)	0 (0%)
Internal resource without governance mechanism	13 (35%)	2 (6%)	3 (18%)	0 (0%)
Mixed objective - Generic carried & independent measurement	3 (8%)	1 (3%)	5 (29%)	0 (0%)
Environmental objective – symbolic structuring	1 (3%)	6 (19%)	0 (0%)	0 (0%)
Environmental objective – robust incentive	0 (0%)	4 (12%)	4 (24%)	0 (0%)
Environmental objective – complete architecture	0 (0%)	8 (25%)	2 (12%)	1 (33%)
Social objective – complete architecture	0 (0%)	7 (22%)	2 (12%)	2 (67%)

Yet, in the data, we do not observe a distribution consistent with this hypothesis. Funds with operational additionality remain dispersed: 8 funds out of 17 (47%) fall under the most robust signatures (*complete architecture or robust incentive*), while 9 out of 17 (53%) are spread across intermediate or weakly coercive signatures (*Minimal IMM Process*, *Internal Resource*, and *Mixed Objective – Generic Carried*). By contrast, 19 funds with financial additionality (59%) display the most robust organizational signatures. This may suggest that the architectures observed reflect in part a **signaling** or contingent logic rather than systematic alignment with the precise nature of the sustainability thesis claimed: even when the additionality strategy is operational, we mainly observe low or intermediate architectures. This also raises the question of how potential trade-offs are managed for these funds displaying operational additionality strategies.

→ **The architectures observed appear only imperfectly aligned with the theoretical logic of the additionality claimed.**

Fifth result: labels, statuses, and regulation do not capture the heterogeneity observed

While regulation and frameworks provide a useful first layer of information, they describe the effective structuring of funds only imperfectly. The final typology reveals several clear gaps between the *ex ante* categories derived from market frameworks and the reality of the architectures observed.

- **11 non-impact funds out of 67 (16.4%)** display a comprehensive architecture around a sustainable objective, whether environmental, social, or mixed.
- **4 Article 9 funds out of 30 (13%)** retain minimal or symbolic structuring, despite a more demanding regulatory positioning.
- **4 Article 8 funds out of 42 (9.5%, rounded to 10%)** display, by contrast, a complete architecture around a sustainable objective.

For industry stakeholders, this is a further indication that these signals remain self-classifications, masking unequal structuring realities.

The first macro partition also conceals substantial internal heterogeneity. The cluster *Structured Sustainability without Dedicated Governance* is distributed across **5 organizational signatures**, while the cluster *Structured Sustainability with Dedicated Governance* is distributed across **6 signatures**. Three intermediate signatures are especially prominent in this gray area: *Internal Resource without Governance Mechanism* (18 funds, 20.2%), *Mixed Objective – EF Carried + Moderate IMM* (9 funds, 10.1%), and *Environmental Objective – Symbolic Structuring* (7 funds, 7.9%).

In other words, behind similar statuses, the effective combinations of objectives, resources, and mechanisms remain very different, especially among Article 8 funds and funds claiming a sustainable approach without fully coercive governance.

$\rightarrow$ **SFDR categories, the distinction between impact and non-impact, and a reading of fund structuring based on the simple presence of organizational mechanisms provide only an imperfect account of the organizational architecture of funds.**

Current limits and directions for using the database

An analysis that remains descriptive

This database already enables the detailed documentation of the structuring of **89 funds** from their commercial and legal documents, but it calls for several interpretive precautions. The results presented here are first and foremost **purely descriptive**: clustering brings out statistical proximities between vehicles, but it establishes **no causal relationship** between the different mechanisms. The correlations observed between signatures, SFDR categories, impact-fund status, or additionality strategies must therefore not be interpreted as effects.

A sample representative of heterogeneity, but not of the market

The sample was constructed on a voluntary basis. It covers 89 funds, including 52 pursuing a sustainable objective and 22 impact funds: this is sufficient to bring out robust regularities, but it cannot be fully representative of the market. Several signatures remain rare, and some configurations appear only for a small number of vehicles. Lastly, the analysis concerns the **organizational structures formalized** in PPMs, fund legal documentation, ESG charters, and side letters: it informs very well on contractual commitments and governance architecture, but not directly on the **actual practices** of investment, monitoring, or day-to-day trade-offs.

Directions

This database opens up several natural avenues that could be explored:

- analysis of **team composition** and its articulation with the signatures observed, and inclusion of the **fundraising date** to document the historical dynamic,
- **matching** with ESG and financial performance data at vehicle level to analyze the effect on performance, and **matching** with the effective composition of portfolios to analyze the effect on investment practices,
- **matching** with disclosure and communication data,
- progressive extension of the survey mechanism at an international scale.

→ **This database is less an endpoint than an empirical foundation for subsequently analyzing, more finely, the real effects of the organizational architectures of funds.**

Appendix

Clustering robustness

This appendix presents the data preparation, decision rules, and robustness tests associated with the macro, micro, and synthesis clusterings.

Data preparation and selected pipeline

The clusterings rely on a **dissimilarity matrix**, that is, a square matrix in which each cell $d(i, j)$ measures the gap between two funds i and j on the basis of the variables retained after data preparation. The diagonal is zero, since each fund is identical to itself, and the matrix is symmetric, since the distance from fund i to fund j is the same as the distance from fund j to fund i. Each cell is constructed by aggregating, variable by variable, the observed gaps between the two funds.

The distance retained is **Gower distance**, because it enables us to work in a single space with binary, ordinal, and quantitative variables. For two funds i and j, we write:

$$d(i, j) = \frac{\sum_h w_{ijh}\, \delta_{ijh}}{\sum_h w_{ijh}}.$$

The term δ_{ijh} denotes the **contribution** of variable h to the dissimilarity between funds i and j. For a symmetric binary variable, this contribution is 0 if the two funds have the same value and 1 otherwise. For an asymmetric binary variable, the same calculation is retained, but the case $(0, 0)$ is not interpreted as an informative similarity. For a quantitative variable, the contribution is the absolute difference between the two funds, scaled by the observed range of the variable. For an ordinal variable, the modality is first repositioned on an ordered scale and then compared as a normalized numerical variable. This co-absence is sometimes informative (for example: no Carried EF mechanism, or no EF Committee). However, because we work on marketing and contractual documentation, we cannot rule out that some GPs implement certain practices (for example: have a sustainability team, use ESG advisory support, etc.) without explicitly mentioning them in their documentation. It is therefore useful not to treat co-absence as informative for these variables. In the following slides, we present the variables considered symmetric (informative co-absence).

The term w_{ijh} indicates whether the contribution of variable h is **valid**. A contribution is valid when the comparison is informative and effectively usable. In practice, $w_{ijh} = 0$ when one of the two values is missing, or when the variable is asymmetric binary and both funds take value 0. Gower distance is therefore the weighted average of valid contributions only. This choice prevents a shared absence of mechanism, very frequent in the database, from artificially bringing two funds closer together.

Clustering is then estimated using **PAM / k-medoids**. Unlike *k-means*, which assumes a Euclidean space and summarizes each group by an average center that is sometimes not observed, *k-medoids* works directly on the distance matrix and anchors each group around a real observation. This method is therefore better suited to mixed data and more robust to atypical observations. It is also useful analytically in our case because it yields one "typical" observation per cluster. The same logic is applied at three levels: a **macro** clustering on the overall architecture of funds, **micro** clusterings by module, and a **synthesis** clustering built from the modalities of the micro modules and the presence of a Sustainability Team.

Decision process and robustness

The choice of the number of clusters retained is based on four criteria: **average silhouette**, **partition stability**, a minimum size for the smallest cluster (in order to avoid having a multitude of clusters with 2 or 3 funds), and the interpretability of the partition. The **silhouette** measures the quality of separation of a partition. For a fund i, let $a(i)$ denote the average distance to its own cluster and $b(i)$ the smallest average distance to another cluster. The silhouette score is then written:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

A silhouette close to 1 indicates that a fund is clearly closer to its own cluster than to the others; a value close to 0 signals a fuzzy boundary; a negative value suggests that assignment to another cluster would be more plausible.

Robustness is assessed through **subsampling**. The principle is to randomly remove part of the sample, re-estimate the clustering, and then compare this new partition to the reference partition. In other words, we do not consider a first clustering credible simply because it is computed on the full sample: we seek to verify that it persists when the sample is reasonably perturbed. For each iteration, we compute the Adjusted Rand Index and the Jaccard index of the smallest cluster, then analyze the distribution of these indicators to assess partition stability.

The **Adjusted Rand Index** compares two partitions in terms of pairs of funds classified together or separately. Let n denote the total number of funds, n_{ij} the number of funds placed in **cluster** i of the first partition and in **cluster** j of the second, a_i the size of cluster i of the first partition, and b_j the size of cluster j of the second. The index is written:

$$ARI = \frac{\sum_{ij}\binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \frac{\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}}{\binom{n}{2}}} = \frac{\text{observed agreement} - \text{expected agreement (chance)}}{\text{maximum agreement} - \text{expected agreement (chance)}}$$

An ARI close to 1 indicates that almost the same partition is recovered; a value close to 0 corresponds to random similarity; a negative value signals strong partition instability.

The **Jaccard** index of the minority cluster measures the overlap between the smallest cluster of the reference partition (A) and its best analogue after re-estimation (B):

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

This indicator complements the ARI, because a partition may remain globally stable while still being fragile in its smallest groups. Values close to 1 are interpreted as a sign of strong stability.

Macro clustering: variables used and choice of K

Macro clustering summarizes the overall degree of fund structuring around sustainability issues. When a variable used in the analysis is derived from a recoding, this is indicated explicitly.

Variable used	Symmetric?
EF Committee	Yes
Carried EF Flag	Yes
Sustainability Team	No
ESG Screening Process	No
ESG Support Process	No
Aspirational ESG Statement	No
ESG Advisory Support	No
IMM: Screening	No
IMM: Objectives	No
IMM: Measurement Process	No
Sustainable Investment Objective Flag (derived variable)	No
Functions Covered (derived variable)	n.a.
Focused (derived variable)	n.a.

Macro robustness is assessed through random and stratified subsampling (removing the same proportion from the clusters of the retained partition) of 80% of the sample without replacement, with $B = 1\,000\,000$ iterations, then through targeted *leave-k-out* tests on the minority cluster with removal of 1, 2, or 5 funds, $B = 1\,000\,000$ iterations.

K	Avg. sil.	Min. cluster size	Max. cluster size
2	0.320	43	46
3	0.329	13	46
4	0.290	13	35
5	0.289	11	35
6	0.293	9	31

Test	Avg. ARI	P10	Median ARI	P90	% >
Random subsampling	0.842	0.564	0.944	1.000	63
Stratified subsampling	0.829	0.560	0.919	1.000	62
LKO – 1	0.593	0.593	0.593	0.593	0.
LKO – 2	0.591	0.590	0.590	0.590	0.
LKO – 3	0.581	0.580	0.580	0.590	0.

We retain the $K = 3$ solution, which has the highest average silhouette. This partition is also robust, with a median ARI of 84.2% on the random sample. The residual instability comes from the smallest cluster (the LKO tests deteriorate rapidly).

Sustainable investment objective module

This module isolates the funds for which an explicit sustainability objective is relevant, then describes how this objective is formulated, contractualized, and linked to impact practices.

Variable used	**Symmetric?**
Impact intentionality	No
IMM Process	Yes
Impact reporting	No
Impact analysis	No
Has Any Target (derived variable)	n.a.
Type of additionality (derived variable)	n.a.
Social objective	No
Environmental objective	No

Stability is assessed through 80% subsampling without replacement, with $B = 1\,000\,000$ iterations, a minimum size threshold defined by $\max(4, \lceil 10\% \times n \rceil)$, and then a filter imposing **median ARI** ≥ 0.55 and **median Jaccard** ≥ 0.50.

K	**Silh.**	**Avg. ARI**	**Median ARI**	**Avg. Jacc.**	**Median Jacc.**
2	0.531	0.777	1.000	0.878	1.000
3	0.310	0.934	1.000	0.932	1.000
4	0.389	0.853	0.966	0.702	1.000
5	0.430	0.773	0.713	0.806	1.000
6	0.521	0.889	0.929	0.891	1.000
7	0.521	0.852	0.865	0.597	0.429

We retain the $K = 3$ solution, because it preserves excellent robustness while still revealing interpretable heterogeneity across types of sustainable objectives. Finer partitions can sometimes improve the silhouette, but at the cost of smaller groups and a less stable reading. The $K = 2$ solution produced a clearer split (silhouette of 0.531) but at the cost of an overly simple interpretation (impact funds vs. non-impact funds). The $K = 3$ solution remains highly robust and allows a finer classification by differentiating social-impact and environmental-impact funds.

Carried EF module

This module describes the precise form of extra-financial incentive mechanisms when Carried EF is present.

Variable used	**Symmetric?**
Outcome-based Carried objectives	Yes
Global Carried objectives	Yes
Weighting by amount invested	Yes
Offsetting across objectives	Yes
Linear Carried EF	No
Carried EF amount	n.a.
Carried EF trigger score	n.a.
Carried EF paid at trigger	n.a.
Full vesting cap for Carried EF	n.a.
Intermediate vesting tier	n.a.

Stability tests are conducted through 80% subsampling without replacement, with $B = 1\,000\,000$ iterations, a minimum size threshold of $\max(4, \lceil 10\% \times n \rceil)$, and the same stability filter based on median ARI and median Jaccard.

K	**Silh.**	**Avg. ARI**	**Median ARI**	**Avg. Jacc.**	**Median Jacc.**
2	0.377	0.937	1.000	0.968	1.000
3	0.375	0.792	0.903	0.847	1.000
4	0.427	0.828	0.839	0.804	1.000
5	0.427	0.736	0.738	0.624	0.600
6	0.441	0.804	0.830	0.742	0.833
7	0.418	0.780	0.787	0.540	0.500

We retain the $K = 4$ solution, because it clearly improves average separation relative to $K = 2$ and $K = 3$ while remaining in an acceptable stability zone. Finer partitions do not bring sufficient gains to justify the additional fragmentation. The $K = 6$ solution offers slightly greater granularity (with a marginally higher silhouette) by isolating "purely linear" versus "tiered" configurations. However, this gain comes with weaker stability of the small clusters (lower average ARI and minority-cluster Jaccard) and substantial redundancy across certain groups, notably linked to secondary modalities (weighting) rather than structural differences. We therefore prefer the $K = 4$ solution, which is more interpretable and more stable.

EF Committee module

This module describes how a dedicated committee intervenes when it exists. The retained variables cover its advisory or executive role, the committee's effective rights, and its composition.

Variable used	Symmetric?
Advisory EF Committee	Yes
Decision: impact KPIs monitored	Yes
Decision: EF KPI targets	Yes
Decision: weighting of impact score	Yes
Decision: change in method & indicators	Yes
Decision: Carried EF payout	Yes
Decision: impact orientation	Yes
Decision: investment eligibility	Yes
Decision: OTI measurement	Yes
Decision: validation of measurement	Yes
Decision: beneficiary organizations	Yes
Committee: LPs	No
Committee: independent members	No

Robustness tests are conducted with 80% subsampling without replacement, $B = 1\,000\,000$ iterations, a minimum size threshold of $\max(4, \lceil 10\% \times n \rceil)$, and the common stability filter.

K	Silh.	Avg. ARI	Median ARI	Avg. Jacc.	Median Jacc.
2	0.468	0.908	1.000	0.940	1.000
3	0.416	0.820	0.783	0.784	0.833
4	0.427	0.719	0.710	0.529	0.286
5	0.376	0.715	0.705	0.676	0.750
6	0.348	0.738	0.737	0.651	0.667
7	0.354	0.694	0.703	0.908	1.000

We retain the $K = 2$ solution. It offers both the best average separation and the best stability, while finer solutions quickly introduce more fragile groups without sufficient interpretive gain. The $K = 3$ solution, which could have been retained given its stability, produces a picture similar to $K = 2$, with one cluster of funds with symbolic committees, one cluster with controlling committees composed only of LPs, and one cluster with controlling committees including independent members.

IMM Process module

This module describes the formalization of Impact Management & Measurement objectives and procedures. The retained variables concern the embedding of objectives in documentation and management incentive mechanisms, the use of frameworks, validations by an external committee, and continuity of the mechanism at fund exit.

Variable used	**Symmetric?**
IMM: Frameworks	No
IMM: Transaction Documentation	No
IMM: EF Committee Validation	No
IMM: ManPack	No
IMM: Independent Measurement	No
IMM: Exit Management	No

As for the other micro modules, stability is assessed through 80% subsampling without replacement, with $B = 1\,000\,000$ iterations, a minimum size threshold of $\max(4, \lceil 10\% \times n \rceil)$, and the common stability filter.

K	**Silh.**	**Avg. ARI**	**Median ARI**	**Avg. Jacc.**	**Median Jacc.**
2	0.411	0.832	1.000	0.894	1.000
3	0.416	0.874	1.000	0.867	1.000
4	0.380	0.728	0.725	0.464	0.333
5	0.373	0.681	0.678	0.535	0.400
6	0.409	0.672	0.668	0.517	0.400
7	0.438	0.701	0.694	0.729	0.833

We retain the $K = 3$ solution. It slightly improves separation relative to $K = 2$ and allows a finer interpretation, while preserving excellent stability, whereas finer solutions make the robustness of the smallest groups much weaker.

ESG contractualization module

This module corresponds to the cross-cutting clustering of ESG contractualization. It seeks to distinguish funds according to the degree of formalization in the fund documentation of exclusions, reporting, risk integration, due diligence, ESG analysis, and obligations imposed on portfolio companies.

Variable used	Symmetric?
Sector exclusions	No
ESG reporting	No
ESG risk integration	No
ESG DD	No
ESG analysis	No
Systematic ESG Process	No
ESG in SHA	No
Share Contractual (derived variable)	n.a.

Stability is assessed through 80% subsampling without replacement, with $B = 1\,000\,000$ iterations, a minimum size threshold of $\max(4, \lceil 10\% \times n \rceil)$, and the common stability filter.

K	Silh.	Avg. ARI	Median ARI	Avg. Jacc.	Median Jacc.
2	0.390	0.758	0.835	0.809	0.885
3	0.359	0.834	0.875	0.750	0.750
4	0.376	0.832	0.916	0.751	0.778
5	0.290	0.697	0.671	0.819	0.875
6	0.291	0.710	0.698	0.359	0.200
7	0.272	0.712	0.708	0.650	0.600

We retain the $K = 3$ solution. It preserves strong stability while revealing richer heterogeneity than at $K = 2$, whereas finer solutions do not sufficiently improve substantive readability relative to the additional complexity.

Synthesis clustering of organizational signatures

The synthesis clustering is built from categorical variables that can each take the retained modalities of the micro modules, together with the presence of a Sustainability Team. It therefore no longer compares raw variables, but rather the use of the different organizational modules (characterized by the previous clustering processes), plus the presence of the Sustainability Team.
We removed the IMM screening and measurement variables because they are almost perfectly correlated with the modalities of the IMM Process mechanism.

Variable used	**Symmetric?**
EF Committee modality	n.a.
Carried EF modality	n.a.
IMM Process modality	n.a.
Sustainable objective modality	n.a.
Sustainability Team	No

Stability is assessed through 80% subsampling without replacement, with $B = 1\,000\,000$ iterations, a minimum size threshold of $\max(4, \lceil 0.05 \times n \rceil)$, and the common stability filter based on median ARI and median Jaccard.

K	**Silh.**	**Avg. ARI**	**Median ARI**	**Avg. Jacc.**	**Median Jacc.**
3	0.310	0.784	0.865	0.792	1.000
4	0.357	0.743	0.725	0.815	1.000
5	0.395	0.813	0.797	0.758	0.846
6	0.383	0.804	0.776	0.859	0.900
7	0.385	0.847	0.849	0.680	0.700
8	0.412	0.878	0.881	0.684	0.833

We retain the $K = 8$ solution. The silhouette continues to improve relative to more aggregated solutions, while the stability of the smallest cluster remains acceptable, unlike what happens when fragmentation is pushed further. A $K = 6$ solution was also conceivable. It simply merges the clusters of funds with a comprehensive architecture (whether they have a social or environmental objective), and the clusters relying on a single internal capability (with or without a Sustainability Team). The $K = 8$ solution therefore provides a more fine-grained interpretation and a better silhouette, while remaining very stable.

Research team: Léo Denis (PhD candidate) supervised by Pr. Nicolas Mottis

Léo Denis (Personal page)

- Graduate of ENS Paris-Saclay, PhD candidate at the Centre de Recherche en Gestion of Ecole Polytechnique (since 2022),
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Pr. Nicolas Mottis (Personal page)

- Professor at Ecole Polytechnique,
- Member of the AMF Climate & Sustainable Finance Commission, Chair of the Scientific Committee of the Finansol-FAIR label and Member of the GREENFIN label Committee, Director of the FIR,
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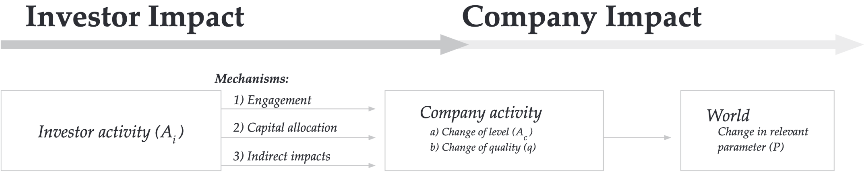

Figure 1: Impact generation chain (Kölbel et al., 2020)

Two main strategies to generate impact:

- Capital allocation: screening techniques and impact investing,
- Shareholder engagement: use of shareholder influence to change the behavior of portfolio companies.

→ Focus on the relationships between "investors" (often considered homogeneous and atomistic) and their portfolio companies.

How should we characterize investment organizations?

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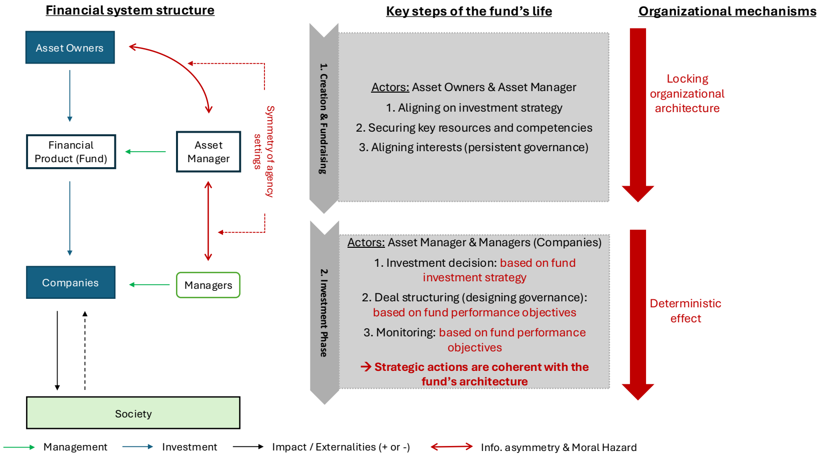

Figure 2: Structuring of investment organizations (Denis & Mottis, 2025)

The key characteristics of fund governance (value-creation objectives, resources to be developed and used, governance mechanisms) are defined and locked in during fundraising, and materialized in the commercial and legal documentation. They do not, in principle, evolve during the active life of the fund, and appear to have a significant influence on management practices (investment and monitoring).

Private Equity: high impact potential and strong intermediation

Growing importance of sustainable finance in this asset class:

- 69% of GPs consider ESG to be a value-creation lever (PwC, 2024),
- Impact or ESG fundraising has been increasing rapidly (approximately 30% p.a. since 2015, McKinsey, 2023), and impact investing reached around $1,164 billion of AUM in 2022 (GIIN, 2023), of which 65% was in Private Equity.

Characterized by **significant impact potential**:

- Access to assets that are virtuous by nature (infrastructure, cleantech, medtech, socialtech, etc.),
- Significant influence on the operations of supported companies (substantial and long-term ownership, proximity to managers), making it possible to encourage the reduction of negative externalities (and the development of positive externalities) of the activity.

Characterized by **complex intermediation between professional organizations**:

- Low liquidity of fund interests, with a long lifespan (10-12 years),
- The *Limited Partnership* form requires *LPs* not to be involved in fund management,
- This induces a complex and explicit structuring of fund governance (decision rights & compensation), historically effective in aligning interests on financial issues (Jensen, 1989).

→ **A highly relevant setting for studying the evolution of the structure of investment organizations alongside the development of sustainable finance**

Appendix — frameworks, sponsoring organization, and functions

Framework	Sponsoring organization	Associated functions
iCI	Initiative Climat International (supported by the PRI)	CMT ; CLIM
UNPRI	Principles for Responsible Investment (PRI)	CMT ; DISC
SDGs	United Nations	GOAL
SASB	IFRS Foundation / ISSB	DISC ; MET
IMP	Impact Management Project	IMM
Operating Principles for Impact Management	IFC / Impact Principles	CMT ; IMM
France Invest & FIR Charter	France Invest + FIR	CMT ; IMM
Paris Agreement	UNFCCC / United Nations	GOAL ; CLIM ; REG
European Taxonomy	European Union / European Commission	CLAS ; DISC ; GOAL ; REG
Greenfin	Ministry for the Ecological Transition	CLAS ; REG
GIIN	Global Impact Investing Network	MET ; DISC
ISR Label	French State / Treasury Directorate	CLAS ; REG
NEC	NEC Initiative	MET ; CLIM
SBTi	Science Based Targets initiative	CLIM ; GOAL
ACT (Ademe)	ADEME + CDP	CLIM ; MET
Business Convention for the Climate	CEC	CMT ; CLIM
FREC	French Government / Ministry for the Ecological Transition	POL ; GOAL
Institut de la Finance Durable Charter	Institut de la Finance Durable	CMT
Impact Frontiers	Impact Frontiers	IMM
2X Challenge	2X Challenge / 2X Global	CMT ; MET
Relance Label	French State / Treasury Directorate	CLAS ; REG
Net Zero Initiative	Carbone 4	CLIM ; GOAL
IPBES	IPBES	POL ; GOAL
LuxFlag ESG	LuxFLAG	CLAS ; REG
GHG Protocol	WRI + WBCSD	MET ; DISC
Global Biodiversity Score	CDC Biodiversite	MET
GRI	Global Reporting Initiative	DISC
OECD Due Diligence Toolkit for Responsible Business	OECD	IMM ; REG
B Corp Assessment	B Lab	CLAS ; MET
CDC ESG Toolkit for Fund Managers	CDC Group / British International Investment	IMM ; DISC